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Road and off road vehicle design

FINAL REPORT OF THE COURSE

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Abstract

This report details a comprehensive series of assignments completed for the Road and Off-Road Vehicles course, using a Lamborghini Miura P400S (1960) as the primary reference vehicle for technical calculations.

The initial phases of the study focus on evaluating the longitudinal dynamics of the reference vehicle, including the determination of optimal gear ratios, acceleration performance, and fuel consumption analysis. This is followed by a consumption and energy analysis of a hybrid vehicle operating under various conditions during a WLTP cycle.

A significant portion of the report investigates vehicle stability and comfort. The handling diagrams are derived for the reference vehicle under varying assumptions for center of gravity (CoG) position, tire pressure, and traction force. Furthermore, a ride comfort analysis is performed using a quarter-car model to compare the discomfort indices of the Alfa Romeo Giulietta, Alfetta, and 159 across different road irregularities.

Structural and system design assignments include computing the fatigue life of an aluminum wheel for a Porsche 718 and the braking system design for a high-performance motorcycle, where deceleration limits such as flipping and adherence are analyzed. The primary engineering task concluded in this report is the kinematic optimization and structural design of the lower arm for a McPherson suspension.

The report concludes with a summary of technical seminars and laboratory visits.

Contents

Abstract	i
Contents	iii
1 General Performance of a Motor Vehicle	1
1.1 Theory	1
1.1.1 Gravitational resistance due to slope	2
1.1.2 Rolling resistance	2
1.1.3 Aerodynamic drag	3
1.1.4 Total resistant force and resistive power	3
1.1.5 Vehicle speed and powertrain relationships	3
1.2 Goals	5
1.2.1 Vehicle selection	5
1.2.2 Identification of vehicle characteristics	5
1.2.3 Estimation of transmission parameters	5
1.2.4 Evaluation of power balance	5
1.2.5 Determination of the highest gear ratio	5
1.2.6 Determination of the lowest gear ratio	6
1.2.7 Determination of intermediate gear ratios	6
1.2.8 Acceleration performance analysis	6
1.2.9 Cases analysis	7
1.3 Reference Vehicle: Lamborghini Miura P400S	8
1.4 Case 1: Vehicle in running order	12
1.4.1 Vehicle Mass	12
1.4.2 Max Speed	12
1.4.3 Highest Gear	13
1.4.4 First Gear	13
1.4.5 Intermediate Gear Ratios	15

1.4.6	Acceleration performance	16
1.5	Case 2: Fully loaded vehicle	18
1.5.1	Effect on Maximum Speed and Fifth Gear	18
1.5.2	Effect on First Gear and Intermediate Ratios	19
1.5.3	Gear Ratios and Acceleration Times	20
1.6	Concluding Remarks	21
2	Road Comfort	23
2.1	Question 1: Most comfortable car and relevant parameters	23
2.1.1	Sensitivity Analysis	24
2.2	Question 2: Reason for the differences	25
3	Consumption Analysis of an Hybrid Vehicle	27
3.0.1	Introduction	27
3.0.2	Procedure	28
3.0.3	Conclusions	30
4	Braking System Design	31
4.1	Introduction and Vehicle Data	31
4.2	Static and Dynamic Load Analysis	32
4.3	Deceleration Limits	32
4.3.1	Flipping Limit	32
4.3.2	Adherence Limit	33
4.4	Braking Force Distribution	33
4.4.1	Only front / rear braking	35
4.5	Aerodynamic Effects at High Speed	36
4.6	Hydraulic System Sizing	36
4.6.1	Master Cylinder and Lever Force	37
4.7	Conclusion	37
5	Suspension Design	39
5.1	Introduction	39
5.2	Objectives	40
5.3	Suspension Kinematics and Optimization	41
5.4	Lower arm forces computation	44
5.5	Material Selection	46
5.6	Finite Element Analysis of the Lower Arm	48
5.7	Suspension Assembly	49

6	Design of Aluminum Wheels	53
6.1	Introduction and Specifications	53
6.2	Bending Moment Calculation	53
6.3	Finite Element Modeling (FEM)	54
6.4	Dynamic Stress Computation	54
6.5	Fatigue Life Estimation	54
6.5.1	Sines Fatigue Criterion	54
6.5.2	Wohler Curve and Cycle Prediction	55
6.6	Conclusion	56
7	Seminars Summary	57
7.1	Visit at Drismi	57
7.1.1	Laboratory Objective	57
7.1.2	Laboratory Procedure	57
7.1.3	Subjective Frequency Response Function (FRF)	57
7.1.4	Acceleration Data	58
7.1.5	Conclusions and Next Steps	60
7.2	Visit at LaST	60
7.3	Cromodora Aluminum Alloy Wheels	61
7.3.1	Introduction	61
7.3.2	Company Overview	61
7.3.3	Aluminum Alloy Wheel Types	61
7.3.4	Technical Features and Specifications	62
7.3.5	Advanced Product Quality Planning (APQP)	62
7.3.6	Learning Outcomes	62
7.3.7	Conclusion	63
7.4	Guest lectures by Capgemini	63
7.4.1	Introduction	63
7.4.2	Functional Safety - Capgemini	63
7.4.3	Key Learning Modules	63
7.4.4	Safety of the Intended Functionality (SOTIF)	64
7.4.5	Practical Application: Active Suspension System	64
7.4.6	FMEA-FMECA Capgemini	65
7.4.7	Risk Assessment and Outcomes	65
7.4.8	Case Study: Front Bumper and ADAS Radar	65
7.4.9	Functional Safety Framework	66
7.5	Visit at Vizzola Pirelli test track	66

7.5.1	Introduction	66
7.5.2	Steering Pad Test	66
7.5.3	Wet Handling Test Drive	67
7.5.4	Dynamics in Low Adhesion	68
7.5.5	Conclusions	68

List of Figures	69
------------------------	-----------

List of Tables	71
-----------------------	-----------

List of Symbols	73
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1 | General Performance of a Motor Vehicle

The general performance of a motor vehicle can be evaluated primarily through the analysis of its longitudinal dynamics, with particular attention to maximum speed, acceleration capability, and gradeability. To this end, a simplified analytical approach is adopted, based on a lumped-parameter representation of the vehicle, which allows a clear understanding of the relationship between vehicle characteristics, resistive forces, and available propulsion power.

1.1. Theory

The vehicle is modelled as a point mass moving along a straight path with longitudinal velocity v and road inclination α . All external forces are assumed to act along the direction of motion and to be applied at the center of mass. Despite its simplicity, this model is sufficiently accurate for evaluating global performance indicators.

Assuming no wheels slip and a rigid vehicle structure, the longitudinal equilibrium of the vehicle is governed by the balance between the traction force generated at the wheels and three resistive forces opposing motion.

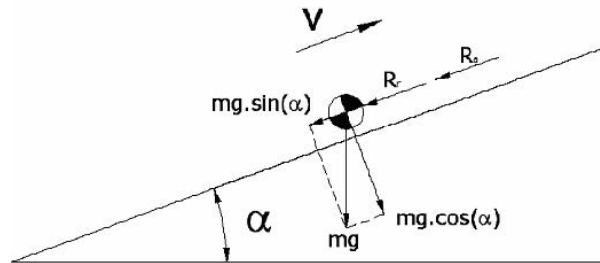


Figure 1.1: Model of vehicle considered as point

1.1.1. Gravitational resistance due to slope

Slope resistance arises from the component of the vehicle weight acting along the direction of motion when the vehicle travels on an inclined road. This force depends exclusively on vehicle mass and road gradient and is independent of speed. It becomes particularly significant when evaluating gradeability and low-speed performance, as it directly increases the traction effort required at the wheels to maintain constant velocity or to accelerate uphill.

$$R_i = mg \cdot \sin(\alpha) \quad (1.1)$$

1.1.2. Rolling resistance

Rolling resistance is mainly associated with energy losses at the tyre–road interface and within the tyre structure, due to deformation and hysteresis phenomena.

$$R_0 = mg \cdot \cos(\alpha) \cdot (f_0 + f_2 v^2) \quad (1.2)$$

The rolling resistance coefficient can be expressed as the sum of a quasi-static term and a speed-dependent contribution:

- The coefficient f_0 represents the quasi-static component of rolling resistance and is mainly related to road surface characteristics and contact patch deformation. This term is approximately constant at low to moderate speeds and depends on tyre type, inflation pressure, and asphalt quality. In this analysis, it is assumed:

$$f_0 = 0.010$$

- The additional term $f_2 v^2$ accounts for velocity-dependent losses, which become relevant at higher speeds. These losses are primarily related to internal damping phenomena within the tyre structure, increased cyclic deformation rates, and, to a minor extent, aerodynamic effects around the rotating wheel. It is assumed:

$$f_2 = 0.5 \cdot 3.62 \times 10^{-6}$$

1.1.3. Aerodynamic drag

Aerodynamic drag is caused by the interaction between the vehicle body and the surrounding airflow. It increases with the square of the vehicle speed and becomes the dominant resistive force at high velocities. Aerodynamic drag primarily limits the maximum achievable speed and strongly affects fuel consumption or energy demand at high speed.

$$R_a = \frac{1}{2} \rho C_x S v^2 \quad (1.3)$$

The magnitude of this force depends on the air density ρ , the vehicle frontal area S , and the drag coefficient C_x , which summarizes the aerodynamic efficiency of the vehicle shape.

1.1.4. Total resistant force and resistive power

By summing the three components, the total resistant force acting on the vehicle is:

$$R = R_i + R_0 + R_a \quad (1.4)$$

Multiplying the total resistant force by the vehicle longitudinal velocity yields the resistive power:

$$N_R = R \cdot v \quad (1.5)$$

1.1.5. Vehicle speed and powertrain relationships

The vehicle longitudinal speed v can be expressed as a function of the engine rotational speed n , measured in revolutions per minute (rpm), and of the transmission and final drive ratios of the powertrain. This relationship can be written as:

$$v = \frac{n 2\pi}{60 \tau_{C_i} \tau_p} r \quad (1.6)$$

where the factor $2\pi/60$ converts the engine speed from rpm to rad/s. The parameter τ_{C_i} represents the transmission ratio corresponding to the i_{th} gear, while τ_p denotes the final drive ratio.

The term r corresponds to the effective rolling radius of the pneumatic tyre, which differs from the nominal unloaded radius r_0 due to elastic deformation under the vertical load.

This effect is taken into account by defining:

$$r = r_0 - \Delta r \quad (1.7)$$

where Δr is the radial deflection of the tyre.

Assuming a linear elastic behaviour of the tyre in the radial direction, the radial deformation Δr can be estimated from the tyre radial stiffness k_t . Under static load conditions, the vertical force acting on each tyre is approximated as one quarter of the vehicle weight, leading to:

$$\Delta r = \frac{mg}{4k_t} \quad (1.8)$$

where m is the total vehicle mass and g is the gravitational acceleration.

Starting from the engine torque characteristic $C_m(\omega)$ and the associated engine power $N_d(\omega)$, the power transmitted to the driving wheels can be evaluated by considering the efficiency of the drivetrain components. The wheel power $N_e(\omega)$ is obtained by scaling the engine power according to the transmission losses, as expressed by:

$$N_e(\omega) = N_d(\omega) \eta_{C_i} \eta_P \quad (1.9)$$

where η_{C_i} is the efficiency of the gearbox corresponding to the selected i -th gear, and η_P is the efficiency of the final drive.

1.2. Goals

Based on the theoretical framework outlined in the previous sections, the following analyses and calculations are carried out in order to evaluate the longitudinal performance of a passenger vehicle and to design the corresponding transmission ratios.

1.2.1. Vehicle selection

A reference car is selected as the case study for the analysis.

1.2.2. Identification of vehicle characteristics

The main vehicle parameters are identified, including vehicle mass, wheelbase, maximum frontal area, aerodynamic drag coefficient, tyre characteristics, and engine performance data. The latter are expressed either in terms of torque $C_m = C_m(n)$ or power $N_d = N_d(n)$ as functions of the engine speed.

1.2.3. Estimation of transmission parameters

The transmission ratios of the final drive τ_p and the individual gearbox ratios are estimated or verified against manufacturer data. In addition, the efficiencies of the gearbox η_{C_i} and the final drive η_p are defined.

1.2.4. Evaluation of power balance

The resistant power and the engine power available at the wheels are calculated and presented as functions of vehicle speed, allowing a direct comparison between tractive and resistive contributions.

1.2.5. Determination of the highest gear ratio

The ratio of the highest gear is determined by varying the corresponding transmission ratio such that the intersection between the tractive power curve and the resistant power curve occurs at, above, or below the maximum desired vehicle speed under level road conditions ($\alpha = 0$).

1.2.6. Determination of the lowest gear ratio

The ratio of the lowest gear is selected to ensure that the vehicle climbs a maximum road gradient of 45%. Operationally, the lowest gear ratio τ_1 is varied in order to shift the engine power curve until it becomes tangent to the resistant power curve corresponding to a 46% slope. The resulting vehicle speed at an engine speed of approximately 1000 rpm is then verified to lie within the range of 7–12 km/h. If this condition is not satisfied, the value of τ_1 is reduced to enforce the minimum allowable vehicle speed.

1.2.7. Determination of intermediate gear ratios

The intermediate gear ratios are determined assuming a geometric progression, so as to ensure an approximately constant variation of engine angular speed during gear shifts. Denoting by n the total number of gear ratios, and by $\tau_{\min}$ and $\tau_{\max}$ the minimum and maximum ratios respectively, the progression is defined as:

$$\tau_t = \frac{\tau_{\min}}{\tau_{\max}} = \text{const} \quad (1.10)$$

For two consecutive gears τ_{i-1} and τ_i , the following relationship holds:

$$\tau_t^{\frac{1}{n-1}} = \frac{\tau_{i-1}}{\tau_i} \quad (1.11)$$

The obtained values are compared with the ratios provided by the manufacturer, and the power curves for each gear are plotted as functions of vehicle speed.

1.2.8. Acceleration performance analysis

The time required to accelerate the vehicle from 70 to 120 km/h in the highest gear is evaluated. Considering that both the engine power and the resistant power depend on vehicle speed, the useful power is defined as:

$$N_u = N_e - N_r \quad (1.12)$$

The corresponding useful tractive force is:

$$R_u = R_e - R_r = m a \quad \Rightarrow \quad a = \frac{R_u}{m} \quad (1.13)$$

The acceleration time is computed by discretizing the speed interval of interest. For each speed increment Δv , assuming a constant average acceleration $\bar{a}$, the incremental time is given by:

$$\Delta t = \frac{\Delta v}{\bar{a}} \quad (1.14)$$

The total acceleration time is then obtained by summing all increments:

$$\Delta t_i = \sum_i \frac{\Delta v_i}{\bar{a}_i} \quad (1.15)$$

1.2.9. Cases analysis

All calculations are performed for two different vehicle loading conditions:

- **Vehicle in running order**, considering:
 - driver only (70 kg);
 - luggage of 5 kg;
 - half tank of fuel.
- **Fully loaded vehicle**, considering:
 - driver and all allowable passengers;
 - luggage of 15 kg per person;
 - full tank of fuel.

1.3. Reference Vehicle: Lamborghini Miura P400S



Figure 1.2: Lamborghini Miura P400S

The Lamborghini Miura was developed in the mid-1960s as an almost revolutionary engineering project, initiated by a team of young engineers at the Sant'Agata Bolognese facility. The concept emerged in 1965 within Lamborghini's technical office under the guidance of Giampaolo Dallara, Paolo Stanzani, and Bob Wallace, who designed a road vehicle based on a competition-derived architecture, featuring a transverse mid-mounted engine and a chassis layout optimized for high-performance dynamics. Initially met with skepticism, the project was eventually approved by Ferruccio Lamborghini after recognizing its innovative potential.

The Miura was officially unveiled at the 1966 Geneva Motor Show and remained in production until 1973, rapidly establishing itself as an automotive icon. Although it was not originally conceived as a factory racing car, the Miura exerted a notable influence on motorsport through experimental variants and competition prototypes developed under Bob Wallace's supervision. These efforts contributed significantly to consolidating Lamborghini's reputation as a manufacturer of high-performance supercars and influenced the subsequent evolution of sports car engineering.

From a mechanical perspective, the Miura P400S is distinguished by its naturally aspirated 60° V12 engine, mounted transversely in a mid-rear configuration, a layout that at the time was almost exclusively adopted in racing vehicles. The engine displacement is 3.9 L, delivering approximately 370 hp at 7,700 rpm and a maximum torque of 387 Nm at 5,500 rpm. The torque and power characteristics are biased toward high engine speeds, enabling optimal performance at elevated vehicle velocities, where aerodynamic drag and

rolling resistance become dominant. This powertrain configuration ensures superior acceleration and sustained high-speed capability compared to most contemporaneous road cars.

The engine is coupled to a five-speed manual transmission integrated within a single housing with the differential. This compact transaxle layout reduces the rotational inertia of the drivetrain, minimizes mechanical power losses, and improves the overall efficiency of power transmission to the rear wheels.

The curb mass of the Miura is 1,298 kg, which is relatively low considering the presence of a V12 engine and the vehicle's performance envelope. Weight reduction was achieved through extensive use of aluminum body panels and a load-bearing monocoque structure that incorporates the engine-transmission assembly as a structural element. The reduced vehicle mass directly lowers inertia, enhances acceleration performance, and mitigates the gravitational component of resistance during uphill operation.

Regarding tyres, the Lamborghini Miura adopts the same dimensions on both axles, namely 205/50 R15, mounted on 15-inch wheels.

From an aerodynamic standpoint, the Miura is characterized by an extremely low and streamlined profile, designed by Marcello Gandini for Bertone. The reduced height and limited frontal area contribute to a decrease in aerodynamic resistance, which becomes the primary opposing force at high speeds. With a drag coefficient $C_x \approx 0.38$ and a frontal area of approximately 1.83 m^2 , the vehicle is capable of achieving speeds close to, and in some conditions exceeding, 280 km/h. Nevertheless, the aerodynamic design philosophies of the era did not emphasize downforce generation, resulting in noticeable front-end lift at very high speeds and a consequent reduction in directional stability.

Overall, the Lamborghini Miura P400S represents an extraordinary synthesis of technical innovation, advanced construction solutions, and design vision. It redefined the concept of the supercar, demonstrating how engineering approaches derived from racing applications could be successfully transferred to a road vehicle, achieving exceptional performance and leaving a lasting legacy in automotive history.

Table 1.1: Powertrain Parameters

Parameter	Symbol	Value	Unit
Maximum Power	$P_{\max}$	370	CV
Maximum Torque	$T_{\max}$	387	Nm
Engine Displacement	V_e	3.9	L
Differential Efficiency	η_P	0.96	–
Gearbox Efficiency	η_{Ci}	0.95	–
1st Gear Ratio	τ_I	2.520:1	–
2nd Gear Ratio	τ_{II}	1.735:1	–
3rd Gear Ratio	τ_{III}	1.225:1	–
4th Gear Ratio	τ_{IV}	1.000:1	–
5th Gear Ratio	τ_V	0.815:1	–
Final Drive Ratio	τ_p	4.08:1	–

Table 1.2: Vehicle Parameters

Parameter	Symbol	Value	Unit
Curb Weight	m_{curb}	1298	kg
Fuel Tank Capacity	V_t	90	L
Wheel Moment of Inertia	J_w	0.6	kg·m ²
Engine Moment of Inertia	J_m	0.4	kg·m ²
Transmission Moment of Inertia	J_t	0.12	kg·m ²
Frontal Area	S	1.83	m ²
Drag Coefficient	C_x	0.38	–
Undeformed Tyre Radius	r_0	0.3334	m
Tyre Radial Stiffness	k_t	200000	N/m

Following the vehicle selection, the next step consists of computing the resistive power of the car under both flat and inclined road conditions, as well as determining the optimal gearbox ratios to ensure satisfactory overall longitudinal performance.

According to the engine map, the Lamborghini Miura P400S is capable of delivering a maximum power of 370 hp at an engine speed of 7,700 rpm and a maximum torque of 387 Nm at 5,500 rpm. These characteristic values define the upper limits of the tractive capability available at the wheels, once drivetrain efficiency losses associated with the gearbox and final drive are taken into account.

The engine torque and power curves provide the basis for evaluating the power transmitted to the driving wheels as a function of engine speed and selected gear ratio. This information is fundamental for comparing the available tractive power with the resistant power required to overcome rolling resistance, aerodynamic drag, and gravitational resistance under different operating conditions.

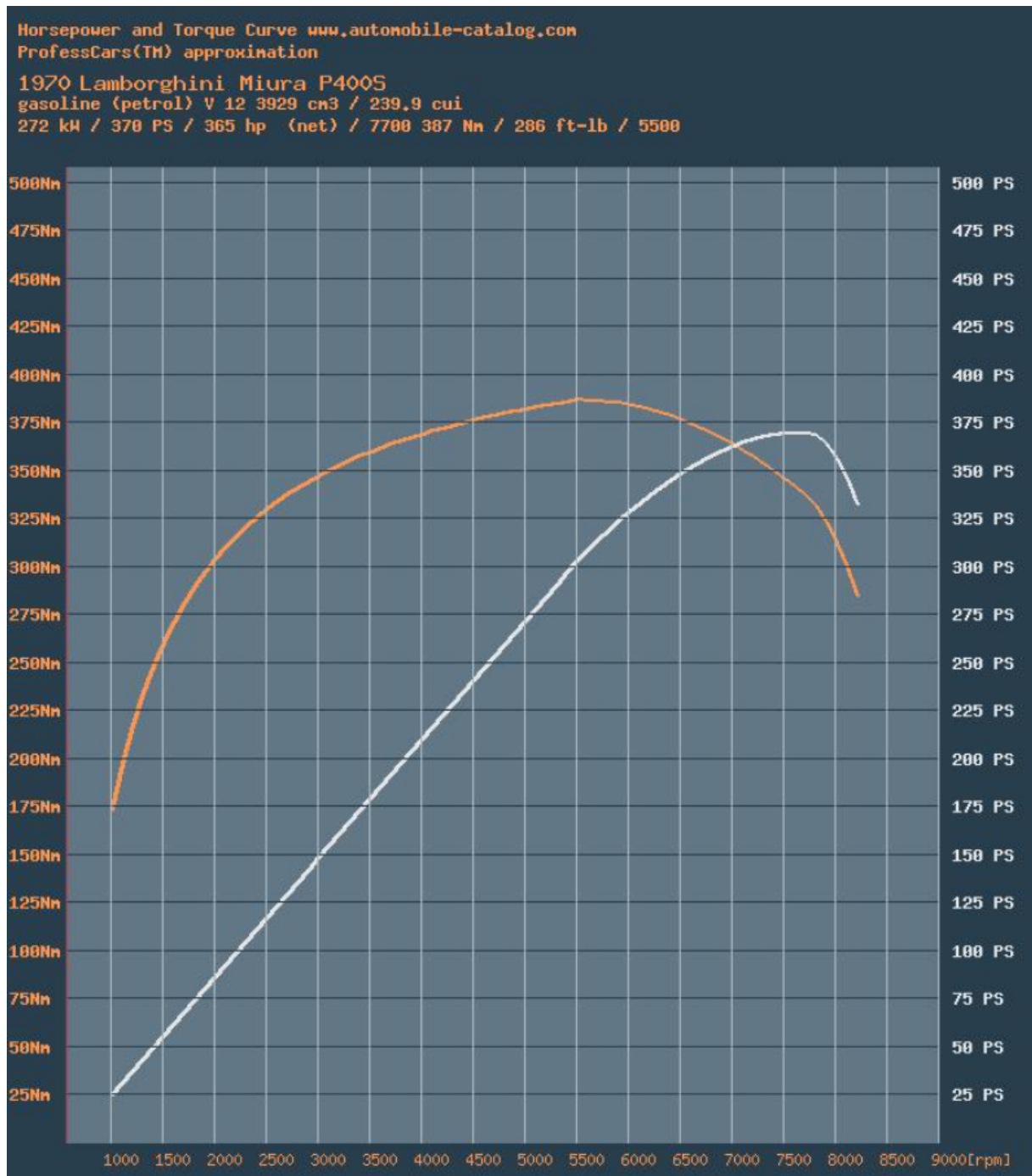


Figure 1.3: Engine torque and power map of the Lamborghini Miura P400S

1.4. Case 1: Vehicle in running order

1.4.1. Vehicle Mass

Initially, the vehicle is analyzed in *running order*, considering the driver, a half-full fuel tank, and a small luggage load of 5 kg. Given the fuel density $\rho_{\text{fuel}} = 740 \text{ kg/m}^3$ and a tank volume $V_{\text{tank}} = 90 \text{ L}$, the mass of fuel in this condition can be calculated as:

$$m_{\text{fuel}} = 0.5 \cdot \rho_{\text{fuel}} \cdot V_{\text{tank}} = 0.5 \cdot 740 \cdot 0.09 = 33.3 \text{ kg} \quad (1.16)$$

Consequently, the total vehicle mass in running order is:

$$m_{\text{running}} = m_{\text{curb}} + m_{\text{driver}} + m_{\text{luggage}} + m_{\text{fuel}} = 1298 + 70 + 5 + 33.3 = 1406.3 \text{ kg} \quad (1.17)$$

With this mass, the resistive forces, including rolling resistance, gravitational resistance (on an incline), and aerodynamic drag, can be computed to evaluate the tractive power required under different operating conditions.

1.4.2. Max Speed

The subsequent analysis focuses on determining gear ratios that optimize longitudinal performance, balancing engine output with resistive forces. After accounting for transmission and differential efficiency losses ($\eta_{\text{gearbox}} = 0.95$, $\eta_{\text{diff}} = 0.96$), the effective power delivered to the wheels is approximately 248,5 kW. Using this available wheel power and the resistive power model, the maximum achievable speed on a level road is estimated as:

$$v_{\text{max}} \approx 279 \text{ km/h}, \quad (1.18)$$

which is fully consistent with the vehicle's stated top speed of 280 km/h.

1.4.3. Highest Gear

The final gear ratio is selected based on the principle of matching the vehicle speed at which the engine delivers its maximum power with the point where the total resistive force intersects the tractive capability curve. By aligning the top gear in this way, the engine operates at or near its peak power output when the vehicle reaches its maximum speed on a flat surface, ensuring optimal exploitation of the available propulsion power. Using this methodology, the calculated ratio for the fifth (top) gear is:

$$\tau_V = 0.808 \quad (1.19)$$



Figure 1.4: Tractive and resistive power for the fifth gear of the Lamborghini Miura P400S

1.4.4. First Gear

The lowest gear ratio is determined to guarantee that the vehicle can climb a maximum road gradient of 45%. In practical terms, this involves adjusting the first gear ratio (τ_1) so that the engine power curve intersects tangentially with the total resistive power curve corresponding to a 46% incline. This ensures that, even under extreme slope conditions, the engine can deliver sufficient tractive effort to maintain motion without stalling.

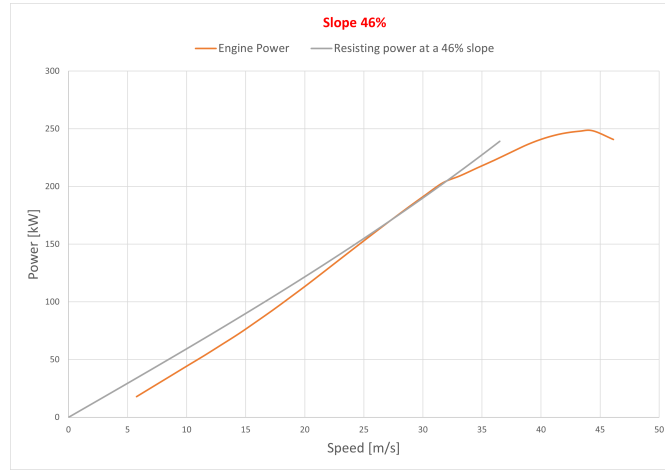


Figure 1.5: Tractive and resistive power at the first gear for maximum incline conditions

Using this initial gear ratio, $\tau_1 = 1.41$, the vehicle is capable of overcoming road gradients of up to 45% or slightly higher. However, selecting the lowest gear purely based on slope requirements can create practical issues in normal driving conditions. Specifically, with the engine idling at 1,000 rpm, the vehicle would travel at a speed that is too high for urban streets, requiring frequent use of the clutch and reducing drivability. Therefore, it is essential to ensure that the vehicle speed at minimum engine rpm falls within a reasonable range, typically between 7 and 12 km/h. For the Miura, a speed of 12 km/h at engine idle was chosen to properly size the first gear, resulting in a first gear ratio of:

$$\tau_1 = 2.439 \quad (1.20)$$

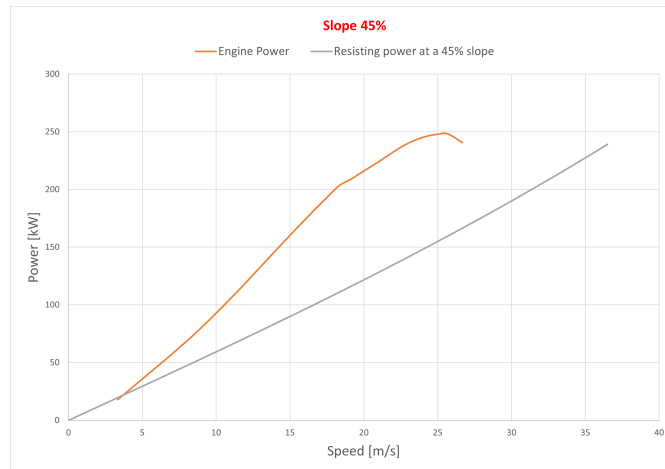


Figure 1.6: Tractive and resistive power at the first gear for maximum incline conditions

1.4.5. Intermediate Gear Ratios

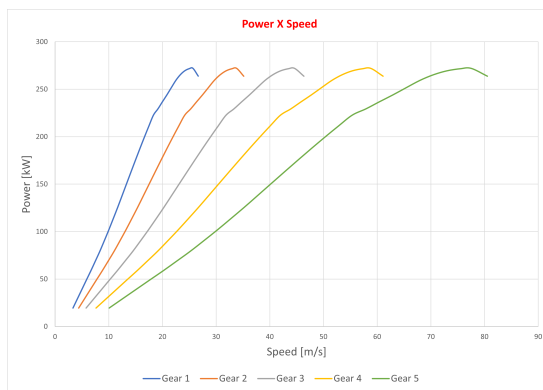
Once the minimum and maximum gear ratios are established, the intermediate ratios can be determined using a geometric progression, which ensures uniform spacing between gear steps, resulting in more consistent engine speed transitions between shifts and a smoother progression of available tractive effort. The geometric progression is defined as:

$$\tau_t = \tau_{\min} \cdot \tau_{\max} = \tau_I \cdot \tau_V \quad (\text{constant factor}) \quad (1.21)$$

$$\frac{\tau_1^{n-1}}{\tau_i - 1} = \frac{\tau_{i-1}}{\tau_i}, \quad \text{where } n = 5 \text{ is the total number of gears.} \quad (1.22)$$

Table 1.3: Calculated and actual gear ratios of the Lamborghini Miura P400S

Gear	Computed ratio	Real ratio
1st	2.439	2.52
2nd	1.850	1.735
3rd	1.404	1.225
4th	1.065	1.000
5th	0.808	0.815



(a) Power x Speed



(b) Speed x Engine RPMs

Figure 1.7: Tractive power on speed and Speed on RPMs for the all the gears of the Lamborghini Miura P400S

The difference between the calculated ratios and the real gear ratios of the Miura's gearbox arises from the fact that calculated ratios, such as those obtained using a geometric progression, are a purely mathematical simplification, whereas the actual ratios were

designed based on practical performance criteria. A geometric progression assumes that the ratio between successive gears is constant, producing uniform steps on a logarithmic scale. While this approach is convenient for theoretical modeling, it does not account for the engine's real torque and power curves, nor does it consider the requirements of actual driving conditions.

The designers of the Miura chose the intermediate gear ratios to optimize real-world acceleration and to keep the engine operating near its peak torque during shifts. The V12 engine of the Miura develops maximum torque at relatively high engine speeds, and if the ratios were strictly geometrically spaced, the engine would drop too far from its optimal torque range after a gear change, causing gaps in acceleration and reducing drivability. Instead, the real ratios were selected to provide smoother acceleration, maintain sufficient torque across gears, and balance performance with control and comfort.

1.4.6. Acceleration performance

The vehicle's acceleration is analyzed in a *rolling start scenario*, meaning that the car begins from a fixed speed and the time required to reach a higher velocity is evaluated while remaining in the same gear. In this study, the focus is on the Miura's acceleration from 70 to 120 km/h in fifth gear.

To perform this analysis, the resistive power at each speed and the corresponding engine rpm required to sustain that speed are first calculated. Then, the maximum engine power available at those rpm is determined. The *useful power* available for acceleration at any instant is given by:

$$N_u = N_e - N_r \quad (1.23)$$

From the useful power, the tractive force acting on the vehicle can be calculated as:

$$R_u = R_e - R_r = m \cdot a \quad \Rightarrow \quad a = \frac{R_u}{m} \quad (1.24)$$

In addition to accelerating the vehicle mass, the rotational inertia of the wheels, transmission, and engine must be accounted for. Therefore, the effective mass is treated as an *equivalent mass*, defined as:

$$m_{\text{eq}} = m_{\text{running}} + \frac{4J_{\text{wheel}}}{r^2} + \frac{J_{\text{trans}}}{r^2 \tau_p^2} + \frac{J_{\text{engine}}}{r^2 \tau_p^2 \tau_{\text{top}}^2} \quad (1.25)$$

The acceleration time is computed by discretizing the speed interval of interest. For each small speed increment Δv , assuming constant average acceleration $\bar{a}$, the incremental time is:

$$\Delta t = \frac{\Delta v}{\bar{a}} \quad (1.26)$$

The total acceleration time is then obtained by summing over all increments across the entire speed range:

$$t_{70-120} = \sum_i \frac{\Delta v_i}{\bar{a}_i} \quad (1.27)$$

Applying this procedure, the calculated acceleration time for the Lamborghini Miura P400S, with 370 hp, is approximately 8.6 s, which is reasonable considering that the Lamborghini Miura P400 (350 hp) accelerates from 80 to 120 km/h in 8.2 s in fifth gear.

1.5. Case 2: Fully loaded vehicle

All the calculations related to gearbox sizing and acceleration performance are repeated for the *full-load condition*. In this configuration, the vehicle is assumed to carry the maximum allowable number of occupants, which in this case corresponds to one passenger in addition to the driver, together with 15 kg of luggage per person and a fully filled fuel tank.

The mass of fuel is computed using the fuel density and tank volume as:

$$m_{\text{fuel}} = \rho_{\text{fuel}} \cdot V_{\text{tank}} = 740 \cdot 0.09 = 66.6 \text{ kg} \quad (1.28)$$

Accordingly, the total vehicle mass in full-load condition is calculated as:

$$m_{\text{full}} = m_{\text{curb}} + m_{\text{driver}} + m_{\text{passenger}} + 2 m_{\text{luggage}} + m_{\text{fuel}} \quad (1.29)$$

$$m_{\text{full}} = 1298 + 70 + 70 + 30 + 66.6 = 1534.6 \text{ kg} \quad (1.30)$$

This increased mass directly affects both the resistive forces and the effective acceleration capability of the vehicle and must therefore be taken into account when evaluating the overall longitudinal performance under fully loaded operating conditions.

1.5.1. Effect on Maximum Speed and Fifth Gear

When the vehicle travels on a flat surface at constant speed, the effect of increased mass on performance is relatively small. Since the vehicle weight does not directly contribute to longitudinal resistive forces under steady-state conditions, its influence mainly arises from the slight increase in rolling resistance, which depends on the normal load acting on the tires. As a result, the maximum achievable speed decreases marginally, from 279 km/h to 277.2 km/h. Consequently, the fifth gear ratio requires only a minor adjustment, changing from 0.808 to 0.809, also accounting for the additional tire deflection caused by the increased vertical load.



Figure 1.8: Tractive and resistive power for the fifth gear of the Lamborghini Miura P400S

1.5.2. Effect on First Gear and Intermediate Ratios

The effects of the increased mass are more noticeable when considering the first gear. In order to maintain an adequate slope of the tractive force curve, the first gear ratio must be slightly increased, resulting in a shorter gear:

$$\tau_1 = 1.51 \quad (1.31)$$

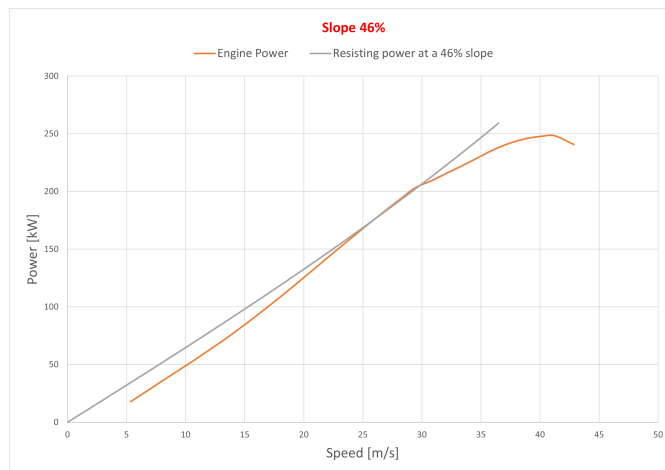


Figure 1.9: Tractive and resistive power at the first gear for maximum incline conditions

However, such a configuration would again be impractical for urban driving. By imposing the same drivability constraint adopted previously, namely a vehicle speed of 12 km/h at minimum engine speed, a revised first gear ratio is obtained:

$$\tau_1 = 2.427 \quad (1.32)$$

which is slightly lower, corresponding to a longer gear due to the increased effective tire deformation. Using this value, the intermediate gear ratios are then recalculated accordingly.

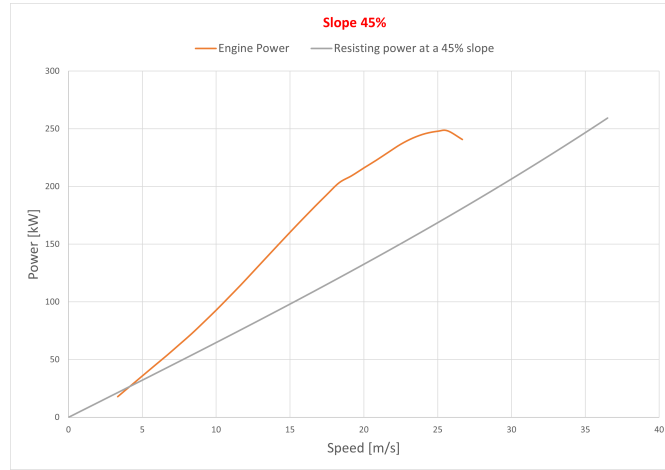


Figure 1.10: Tractive and resistive power at the first gear for maximum incline conditions

1.5.3. Gear Ratios and Acceleration Times

Table 1.4: Gear Ratios in Full-Load and Running Order Condition compared to real ones

Gear	Computed ratio (FL)	Computed ratio (RO)	Real ratio
1st	2.427	2.439	2.52
2nd	1.844	1.850	1.735
3rd	1.401	1.404	1.225
4th	1.065	1.065	1.000
5th	0.809	0.808	0.815

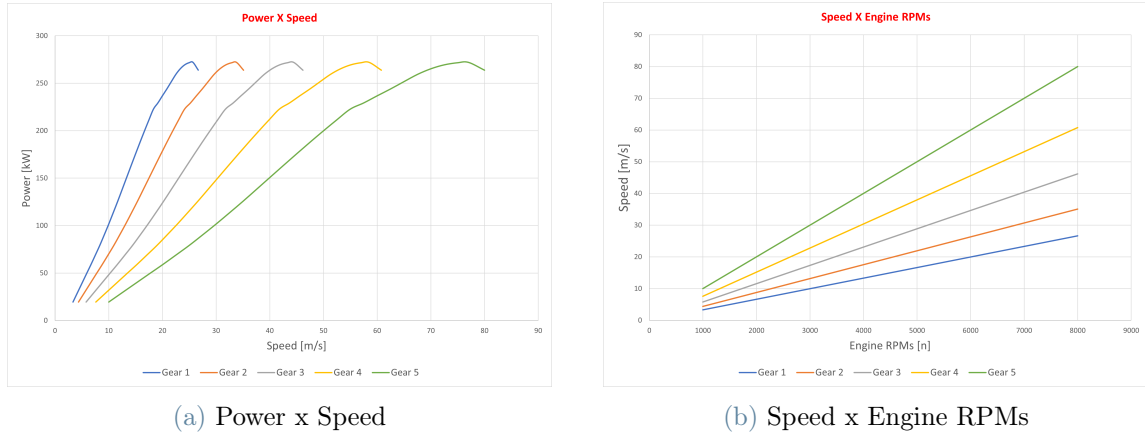


Figure 1.11: Tractive power on speed and Speed on RPMs for the all the gears of the Lamborghini Miura P400S

Vehicle acceleration is also affected by the increased mass. In the full-load configuration, the Lamborghini Miura P400S requires approximately 9.32 s to accelerate from 70 to 120 km/h in fifth gear, compared to 8.6 s in the lighter running-order condition.

1.6. Concluding Remarks

Despite the high power output of the Lamborghini Miura's engine, the increase in vehicle mass leads to measurable, albeit limited, changes in performance. This behavior is typical of high-performance sports cars, whose dynamic response is particularly sensitive to variations in mass, especially in acceleration and low-speed tractive conditions.

2 | Road Comfort

2.1. Question 1: Most comfortable car and relevant parameters

The objective was to determine which car among the Alfa Romeo Giulietta (1960), Alfetta (1984), and 159 (2007) is the most comfortable. This investigation highlights how car comfort has evolved over approximately 60 years. A quarter car model (Figure 2.1) was utilized for this evaluation, using the parameters for damping, mass, and stiffness reported in Table 2.1.

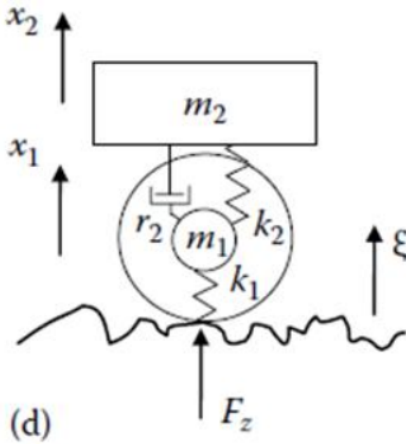


Figure 2.1: Quarter car model scheme.

Table 2.1: Cars data for quarter vehicle model

	Giulietta	Alfetta	159
m_1 [kg]	25	30	33
m_2 [kg]	200	270	350
k_1 [N/m]	100000	150000	270000
k_2 [N/m]	10000	20000	28000
r_2 [Ns/m]	1500	2500	5200

The discomfort index ($\sigma_{\ddot{x}}$) was computed using the following formulas:

$$\sigma_{\ddot{x}}^2 = \frac{1}{2} A_b v \bar{\sigma}_{x2}^2$$

$$\bar{\sigma}_{x2}^2 = \frac{1}{m_2^2} \left(\frac{(m_2 + m_1) k_2^2}{r_2} + k_1 r_2 \right)$$

where A_b represents road irregularities (1.4×10^{-5} m for normal roads; 1.0×10^{-4} m for rough roads) and v is the vehicle speed (30 m/s).

Table 2.2: σ_{x2} for normal and rough roads

	Giulietta	Alfetta	159
σ_{x2} (normal road)	0.930726	1.103865	1.582987
σ_{x2} (rough road)	2.487469	2.950204	4.230711

As shown in Table 2.2, the Giulietta is the most comfortable car, followed by the Alfetta, and finally the 159.

2.1.1. Sensitivity Analysis

A sensitivity analysis was performed on the Alfetta to understand this behavior. Results (Figure 2.2) indicate that the sprung mass (m_2) is the most influential parameter; an increase in m_2 causes a reduction in discomfort (σ_{x2}). Despite the increase in mass in more recent models, the discomfort index rises. This is explained by the significant percentage variation of tire radial stiffness (k_1) and damping (r_2) across the cars, which causes σ_{x2} to increase.

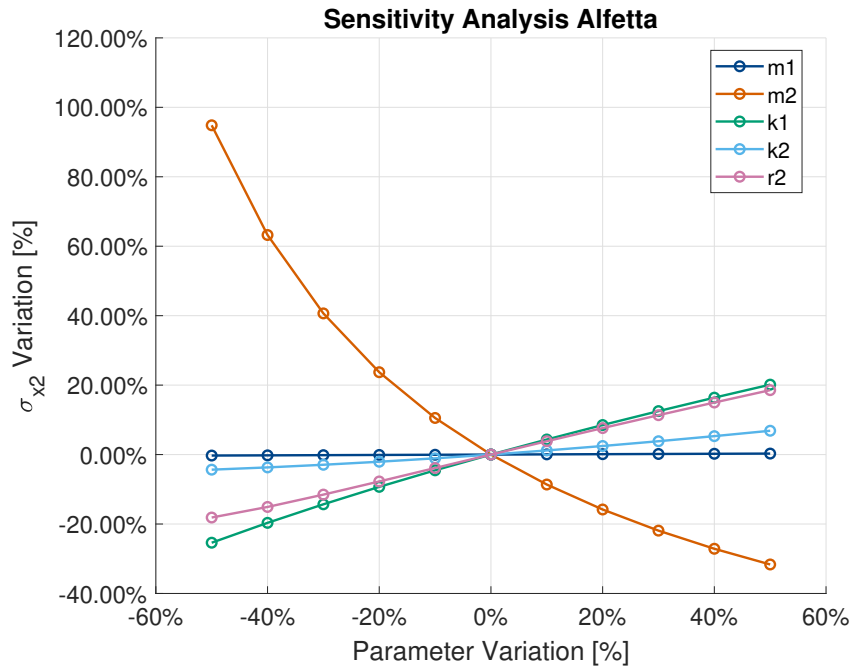


Figure 2.2: Sensitivity for Alfa Romeo Alfetta.

Table 2.3: Parameters' effect on σ_{x2}

	Giulietta	Alfetta	159
m_1	0%	20%	32%
m_2	0%	35%	75%
k_1	0%	50%	170%
k_2	0%	100%	180%
r_2	0%	67%	247%

2.2. Question 2: Reason for the differences

The reasons for the observed evolution in discomfort are as follows:

- **Road Conditions:** Modern roads are predominantly paved, making gravel or dirt road driving (common in the 1960s) unlikely. Better road quality allows for a higher discomfort index than was previously acceptable.
- **Vehicle Characteristics:** Modern cars are faster, heavier, and more dynamically capable.
- **Suspension Stiffness:** A stiffer suspension system is now required to ensure stability, controllability, and safety while sustaining increased vehicle weights. This results in the lack of body roll in corners common in older cars.

3 | Consumption Analysis of an Hybrid Vehicle

3.0.1. Introduction

In this assignment, the vehicle studied is a plug-in Hybrid Electric Vehicle (PHEV) with four operating modes: ICE drive, Electric drive, Hybrid drive, and Energy recovery. The architecture features a front ICE and a rear EM.

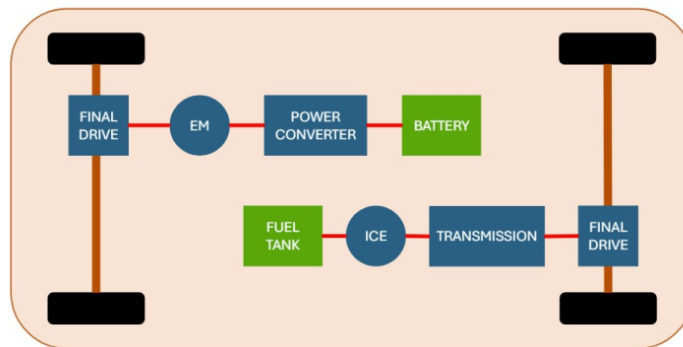
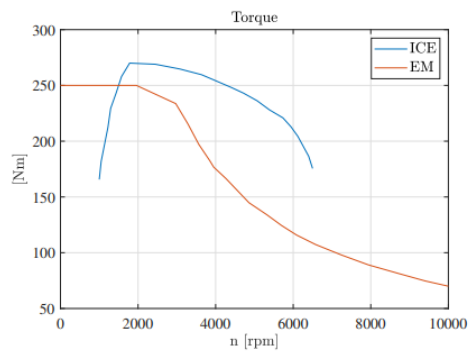
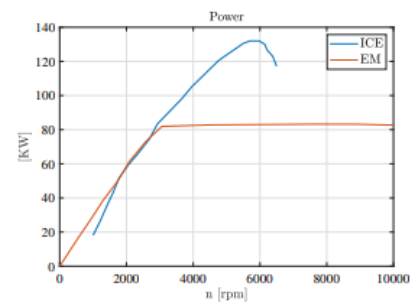


Figure 3.1: Vehicle architecture.



(a) Torque curves.



(b) Power curves.

Table 3.1: Vehicle data

Curb weight (without driver)	1770 kg
$C_x \cdot S$	0.913 m^2
Tyres	235/45 R19
Traction	4x4 front drive with gasoline engine, rear drive with electric motor
ICE type	spark-ignition (line 4, gasoline)
ICE specs	132 kW @5800rpm / 270 Nm (1000 rpm – 6500 rpm)
Engine displacement	1332 cm^3
EM specs	45 kW / 250 Nm
Battery capacity	11.40 kWh
Number of gears	6
Gear ratios	4.459, 2.508, 1.555, 1.142, 0.851, 0.672
Final drive ratio front	3.683
Overall drive ratio rear	8
Regeneration efficiency	0.3

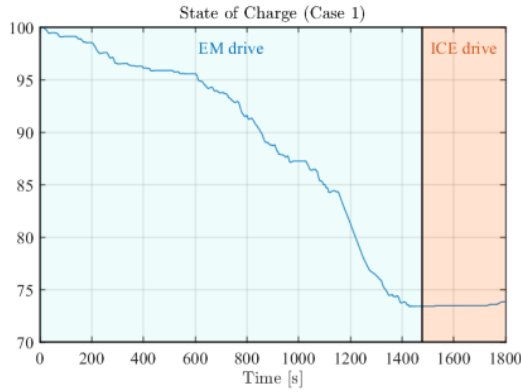
What we have to do is to compute the energy consumption and the fuel consumption during a WLTP cycle (see Assignment 2 for the cycle speed profile), showing also the evolution of the State of Charge (SOC) of the battery. We have to analyse two cases, as in Assignment 1: low load and full load (see the data from Assignment 1). During the WLTP we use only the EM for the first part of WLTP cycle, in particular for low, medium and high phases. For the extra-high step we investigate two possibilities: ICE drive mode with variable power and hybrid drive mode with the ICE operating in the region of maximum efficiency

3.0.2. Procedure

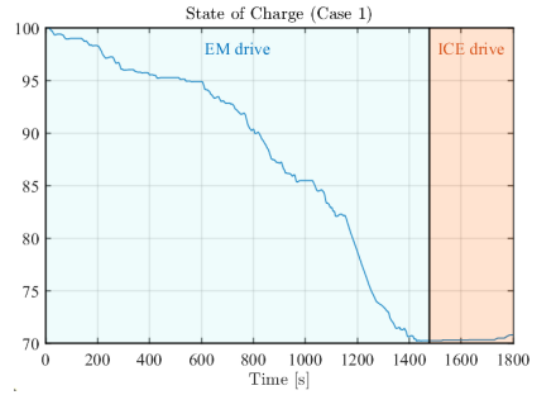
We analyze two cases for the extra-high step of the WLTP cycle: variable ICE power (Case 1) and constant ICE power with CVT (Case 2).

Case 1: Variable ICE Power

In Case 1, the battery discharge is higher under full load, reaching 70% capacity compared to 73% at low load.



(c) Low Load.



(d) Full Load.

Table 3.2: Overall consumption - Case 1

	Fuel cons [Kg]	Fuel cons [l]	Energy cons [KWh]
Low load	0.5826	0.7873	2.9837
Full load	0.7881	1.065	3.3308

Case 2: Constant Power

In Case 2, we fix the working point at 2650 [rpm] where efficiency is maximum (0.35).

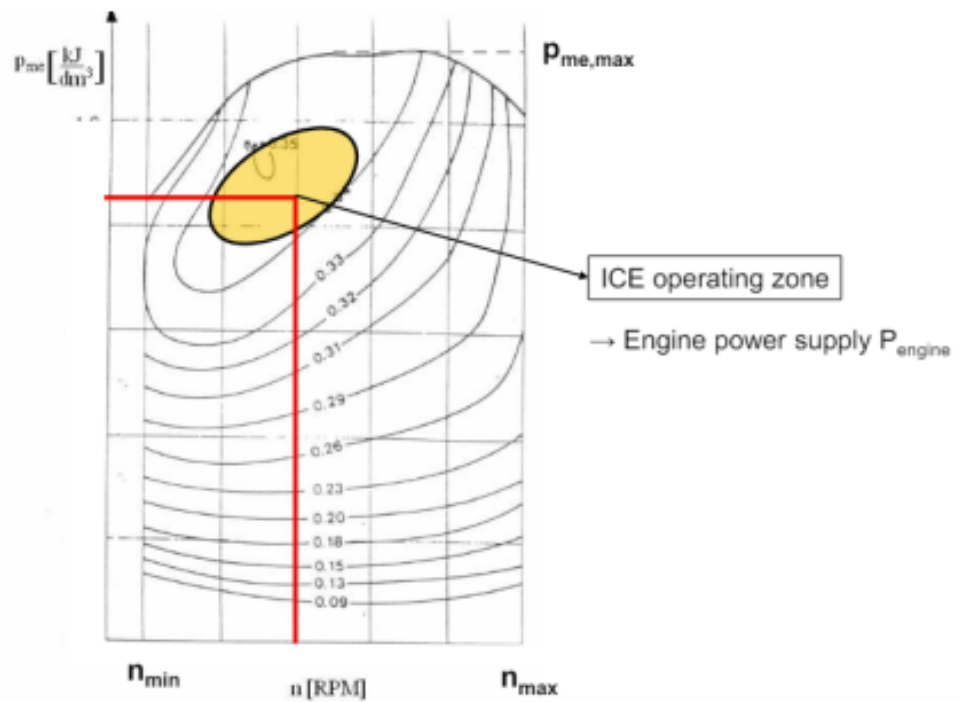
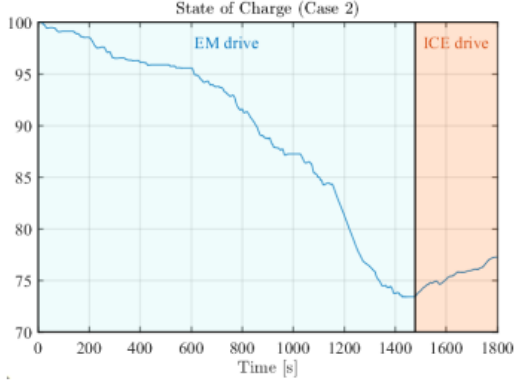
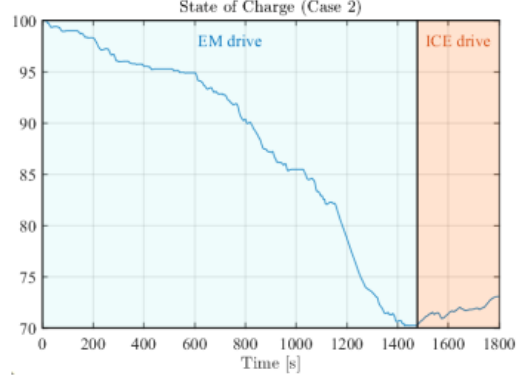


Figure 3.2: Vehicle architecture.

Talking about the SOC we have to look at Fig.3.8 and Fig.3.14. In this case we have that in ICE drive region we have battery charge, recovering 2%-3% of battery in low load condition and 1%-2% in full load.



(a) Low Load.



(b) Full Load.

The results in terms of consumption are collected in Table 3.3. As before in full load condition we have higher Energy consumption pretty the same in fuel consumption. The difference from case 1 is that we have an higher consumption of fuel because part of the fuel energy is used to charge the battery. As a consequence the overall energy consumption is lower in case 2.

Table 3.3: Overall consumption - Case 2

	Fuel cons [Kg]	Fuel cons [l]	Energy cons [KWh]
Low load	0.8492	1.1476	2.5954
Full load	0.8428	1.1389	3.0717

3.0.3. Conclusions

In Case 1, full load results in higher consumption of both fuel and energy. In Case 2, fuel consumption is nearly identical across loads because the working point is fixed, but Case 2 achieves lower overall energy consumption as fuel energy is also used to charge the battery.

4 | Braking System Design

4.1. Introduction and Vehicle Data

The objective of this section is to design and analyze the braking performance of a high-performance motorcycle. The analysis focuses on determining the physical limits of deceleration (flipping vs. adherence) and sizing the hydraulic components required to achieve these limits. The vehicle parameters used for all calculations are summarized in Table 4.1.

Table 4.1: Vehicle and Rider Specifications

Parameter	Symbol	Value
Wheelbase	p	1435 mm
CoG distance from rear axle	b	665 mm
CoG height	h	677 mm
Bike Mass	m_{bike}	157 kg
Rider Mass	m_{rider}	70 kg
Total Mass	m	227 kg
Road Friction Coefficient	μ	0.7

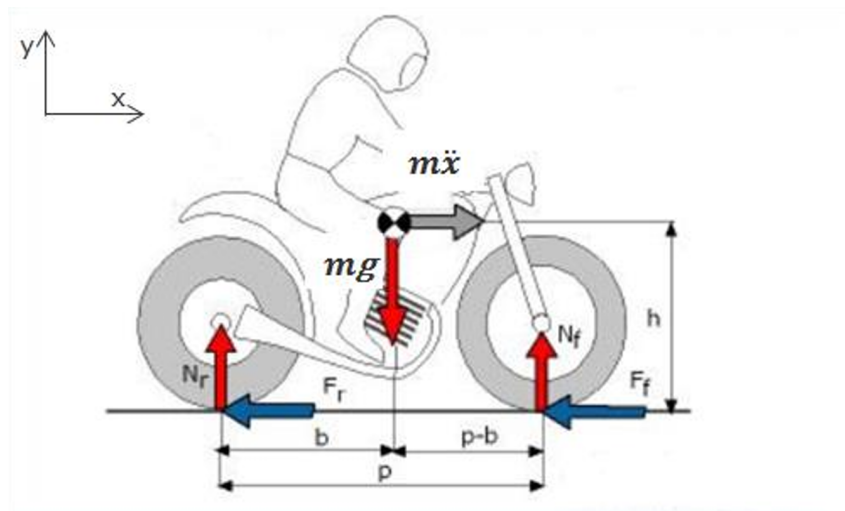


Figure 4.1: Motorcycle scheme.

4.2. Static and Dynamic Load Analysis

Before analyzing the deceleration, we establish the static weight distribution. The vertical loads on the front (N_f) and rear (N_r) wheels are calculated through moment equilibrium around the rear contact point to find $F_{z,f}$:

$$F_{z,f} \cdot p - mg \cdot b = 0 \quad (4.1)$$

Doing the same equilibrium at the front contact point 2 equations are obtained describing the static distribution:

$$F_{z,f} = mg \cdot \frac{b}{p} = 1024.36 [N] \quad (4.2)$$

$$F_{z,r} = mg \cdot \frac{p-b}{p} = 1202.51 [N] \quad (4.3)$$

The resulting distribution is 46% front and 54% rear

4.3. Deceleration Limits

Two physical limits constrain the maximum deceleration of the motorcycle: the geometric flipping limit and the tire-road adherence limit.

4.3.1. Flipping Limit

In this section the aerodynamic effects are not taken into account the flipping limit occurs when the rear wheel loses contact with the ground ($F_{z,r} = 0$). Neglecting aerodynamic effects, this limit is purely geometry-dependent. We have to perform a moment equilibrium around the front tire considering also the inertia contribution:

$$F_{z,r} \cdot p - mg \cdot (p-b) + ma_x \cdot h = 0 \quad (4.4)$$

From this equation $F_{z,r}$ is isolated and then putting it equal to 0 we derive the acceleration for flipping:

$$a_{x,flip} \leq g \cdot \frac{p-b}{h} = 1.14g \quad (4.5)$$

4.3.2. Adherence Limit

The adherence limit is reached when the tires lose grip. For a friction coefficient $\mu = 0.7$, the maximum deceleration is:

$$\begin{cases} F_{z,f} = mg \cdot \frac{b}{p} + ma_x \cdot \frac{h}{p} \\ F_{z,r} = mg \cdot \frac{p-b}{p} - ma_x \cdot \frac{h}{p} \end{cases} \quad (4.6)$$

We also now that:

$$\begin{cases} ma_x = F_{x,f} + F_{x,r} \\ F_{x,f} = \mu \cdot F_{z,f} \\ F_{x,r} = \mu \cdot F_{z,r} \end{cases} \quad (4.7)$$

Putting together all the information collected before we can write:

$$ma_x = \mu \cdot \left(mg \cdot \frac{b}{p} + ma_x \cdot \frac{h}{p} \right) + \mu \cdot \left(mg \cdot \frac{p-b}{p} - ma_x \cdot \frac{h}{p} \right) \quad (4.8)$$

The resulting inequality that describes the adherence limit is:

$$a_{x,slip} \leq \mu \cdot g = 0.7g \quad (4.9)$$

Since $a_{x,slip} < a_{x,flip}$ in this specific exercise, the vehicle is limited by adherence rather than geometry.

4.4. Braking Force Distribution

To achieve the maximum deceleration of $0.7g$, the braking force must be distributed to match the dynamic load on each wheel. The optimal repartition is calculated as:

$$\begin{aligned} F_{x,f} &= \mu \cdot \left(mg \cdot \frac{b}{p} + ma_x \cdot \frac{h}{p} \right) = 1237.16[N] \\ F_{x,r} &= \mu \cdot \left(mg \cdot \frac{p-b}{p} - ma_x \cdot \frac{h}{p} \right) = 321.65[N] \end{aligned} \quad (4.10)$$

Summing these two values we can achieve the totale force ($F_{x,tot} = F_{x,f} + F_{x,r} = 1558.81[N]$).

Now we can determine the optimal distribution just performing the ratio between the obtained values:

$$\begin{aligned}\frac{F_{x,f}}{F_{x,tot}} &= 0.79[-] \\ \frac{F_{x,r}}{F_{x,tot}} &= 0.21[-]\end{aligned}\tag{4.11}$$

In the following images we have a graphical representation of the values we have obtained. We can notice from the previous equations, during the deceleration we have load transfer and as a consequence the available braking force at the front is higher than the one at the rear (as we can notice from the $F_{x,f}$ - $F_{x,r}$ plot).

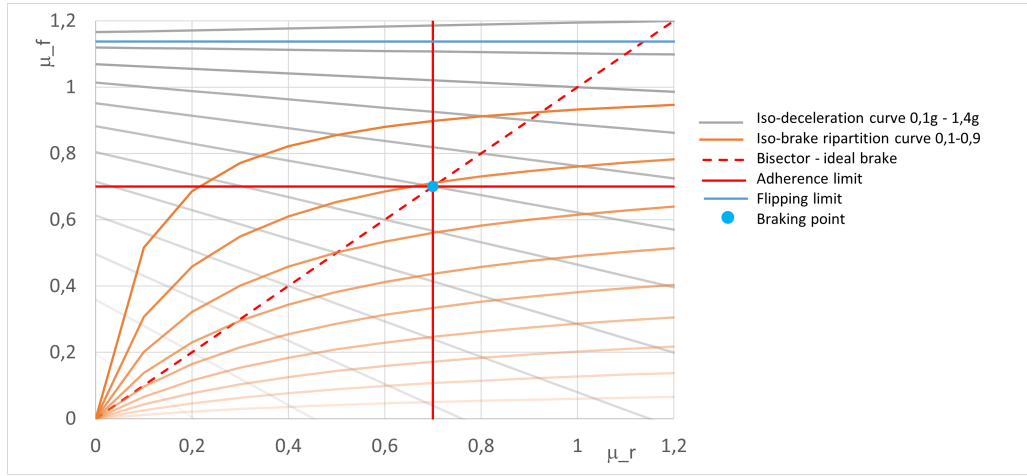


Figure 4.2: $\mu_r - \mu_f$ diagram.

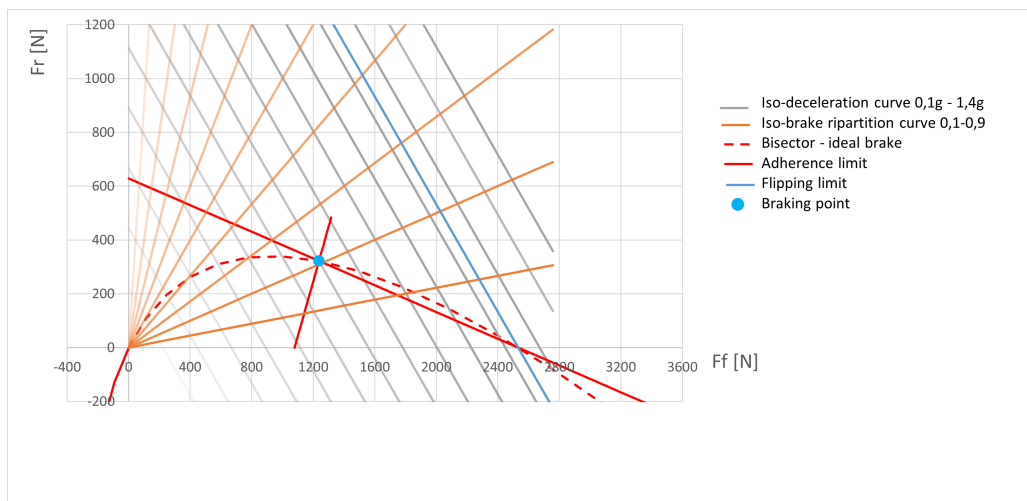


Figure 4.3: $F_{x,f} - F_{x,r}$ diagram.

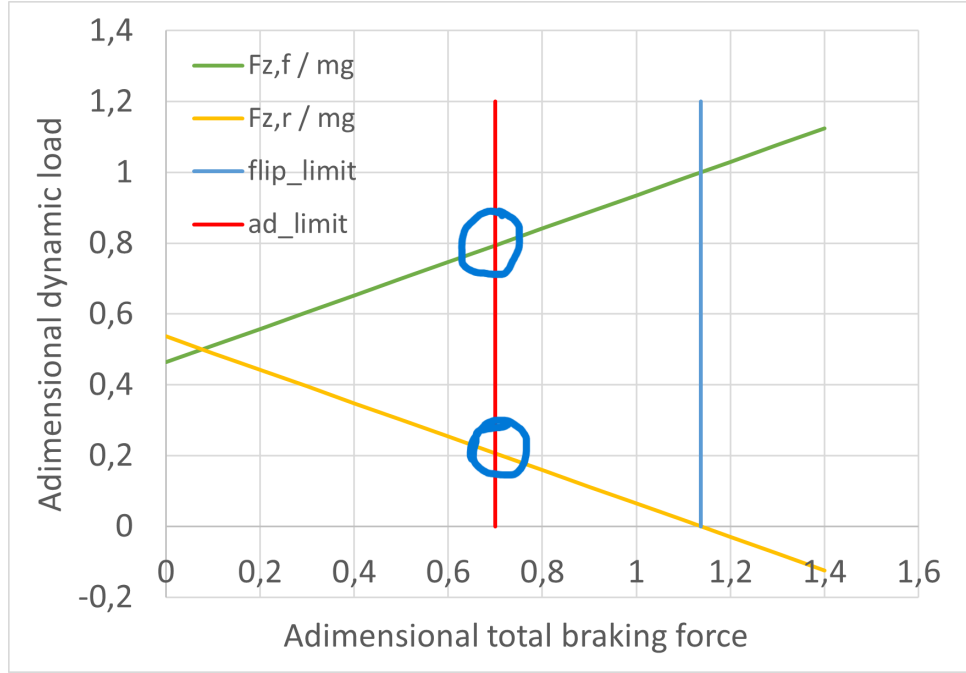


Figure 4.4: Deceleration limits.

4.4.1. Only front / rear braking

Now we investigate what happens if the rider try to brake using only the front or only the rear brake. Starting from full front braking, we can write:

$$ma_x = F_{x,f} = \mu \cdot \left(mg \cdot \frac{b}{p} + ma_x \cdot \frac{h}{p} \right) \quad (4.12)$$

From which we can derive:

$$a_{x,allfront} = \frac{\mu \frac{b}{p}}{1 - \mu \frac{h}{p}} \cdot g = 0.48g \quad (4.13)$$

We repeat the same operation for the only rear braking and we star from:

$$ma_x = F_{x,r} = \mu \cdot \left(mg \cdot \frac{p-b}{p} - ma_x \cdot \frac{h}{p} \right) \quad (4.14)$$

And then we can calculate the acceleration as we done before:

$$a_{x,allfront} = \frac{\mu \frac{p-b}{p}}{1 + \mu \frac{h}{p}} \cdot g = 0.28 \quad (4.15)$$

As we can expect both the cases are worse than the optimal one; as a consequence of load transfer in braking we have that if we use only the front disk the deceleration is still close to the optimal one, instead in the case of full rear braking the value obtained is much lower.

4.5. Aerodynamic Effects at High Speed

At high speeds ($v = 315$ km/h), the aerodynamic drag force (F_D) contributes to the deceleration and modifies the flipping limit. The drag force is:

$$F_D = \frac{1}{2} \cdot \rho \cdot A \cdot C_D \cdot V^2 = 1102.50[N] \quad (4.16)$$

With $\rho = 1.2 \text{ kg/m}^3$, $C_d = 0.6$, and $A = 0.4 \text{ m}^2$, the drag force provides an additional decelerating component and increases the effective flipping limit:

The drag force is applied in the center of pressure (CoP). The text does not give us information about the position of that point; we decide to consider a CoP at the same height of the CoG. The drag force can be written as:

The moments equilibrium around the front tire is:

$$F_{z,r} \cdot p + m a_x \cdot h - m g \cdot (p - b) - F_D \cdot h \quad (4.17)$$

From which we derive:

$$a_x = \frac{m g \cdot (p - b) + F_D \cdot h}{m \cdot h} = 1.63 \quad (4.18)$$

As we can expect the drag force help us to stop the vehicle; the result is an increase of the maximum deceleration from $1.14g$ to $1.63g$.

4.6. Hydraulic System Sizing

The front braking system consists of two 340 mm discs and 36/38 mm calipers (M4 configuration). and we consider $\mu_{disk-pad} = 0.5$ (disc/pad friction coefficient). We can start determining the $F_{caliper}$ as:

$$F_{caliper} \cdot R_{eff} \cdot n_{calipers} = F_x \cdot R_{wheel} \quad (4.19)$$

Where R_{eff} is the effective radius of the disk, R_{wheel} is the wheel diameter, F_x is the longitudinal force at the contact point and $F_{caliper}$ is the force at the disk. Solving for $F_{caliper}$ we obtain 4886.4 [N]. Now we can compute the pressure, knowing the cylinders diameters:

$$p = \frac{F_{caliper}/2}{\frac{\pi \cdot (D_1^2 + D_2^2)}{4}} = 11.4[bar] \quad (4.20)$$

Where $D_1 = 36[mm]$ and $D_2 = 38[mm]$.

4.6.1. Master Cylinder and Lever Force

We compared two Master Cylinders: 19×18 and 19×20 . The rider hand force (F_{rider}) is a function of the mechanical ratio and the hydraulic ratio:

$$F_{rider} = \frac{p \cdot A_{M/C} + K_{flott} \cdot (l_0 - l_{preload})}{(a/i)} + 10\% \quad (4.21)$$

where a/i is the mechanical leverage. In our analysis, we considered:

- Finger position: 100 mm from the fulcrum.
- 10% internal friction losses.
- Internal spring preload.

With the 19x18 cylinder the $F_{rider} = 68.6[N]$ with the 19x20 cylinder the $F_{rider} = 76.2[N]$ Switching from an 18 mm to a 20 mm interax increases the required F_{rider} but results in a shorter lever stroke, providing a "firmer" feel favored by some riders.

4.7. Conclusion

The analysis demonstrates that at low speeds, the motorcycle is adherence-limited ($\mu = 0.7$), while at racing speeds, aerodynamics significantly increase the stability against flipping. The sizing of the 36/38 caliper with a 19x18 M/C ensures that the flipping limit can be reached with a hand force well within the ergonomic limits of a professional rider.

5 | Suspension Design

5.1. Introduction

The **MacPherson suspension** is one of the most widely used independent suspension systems in modern vehicles. It was first developed in the early 1940s by **Earle S. MacPherson**, an American automotive engineer working for General Motors. Originally designed to simplify front-wheel-drive layouts, the system has since become a standard in passenger cars worldwide due to its simplicity, compactness, and cost-effectiveness.

The MacPherson suspension is characterized by a **shock absorber rigidly connected to the hub carrier**, typically accompanied by a helical spring that is not coaxial with the shock absorber. The hub carrier is attached at its lower end to a single lower control arm, often pseudo-triangular in shape, which allows the wheel to move freely in the vertical direction while maintaining lateral stability.

A key feature of the design is the **bearing at the top of the shock absorber**, which allows the strut to rotate smoothly during suspension compression and extension. This arrangement ensures proper wheel alignment and enables the system to respond effectively to road irregularities, improving both comfort and handling.

The main advantage of the MacPherson suspension is its **simple and compact design**, which reduces manufacturing costs and allows for better packaging in vehicles with limited space, particularly front-engine, front-wheel-drive cars. Additionally, having fewer components than double-wishbone suspensions reduces weight and simplifies maintenance.

However, the system has some inherent limitations. The most notable is the **variation of the camber angle** during suspension travel and body roll. When the suspension compresses, the camber angle becomes more negative, causing the wheels to tilt inward. This can affect tire contact with the road, traction, and steering precision, especially when cornering or driving over uneven surfaces. Furthermore, the **track width**, or the lateral distance between wheels, may vary slightly during suspension movement, potentially impacting vehicle stability under dynamic conditions.

In summary, the MacPherson suspension represents an **efficient compromise** between simplicity, cost, and performance. While it may not provide the same high-performance handling characteristics as more complex suspension systems, it remains a practical and reliable solution for the vast majority of passenger vehicles.

5.2. Objectives

The aim of this assignment is to design the overall assembly of a MacPherson strut for a front-drive axle, with particular focus on the suspension lower arm. Due to time constraints, only the main aspects of the suspension design process are considered. It is assumed that a dynamic study of the vehicle has been previously conducted by third parties, which defined specific target values for the suspension's characteristic angles. Additionally, it is assumed that the vehicle layout imposes constraints on the suspension attachment points and linkages.

The main goal is to achieve values as close as possible to the target characteristic angles within these constraints, supporting the design process with a static analysis of the system and a finite element analysis (FEA) of the lower arm. Finally, the design of the lower arm and the full suspension assembly has been presented with technical drawings.

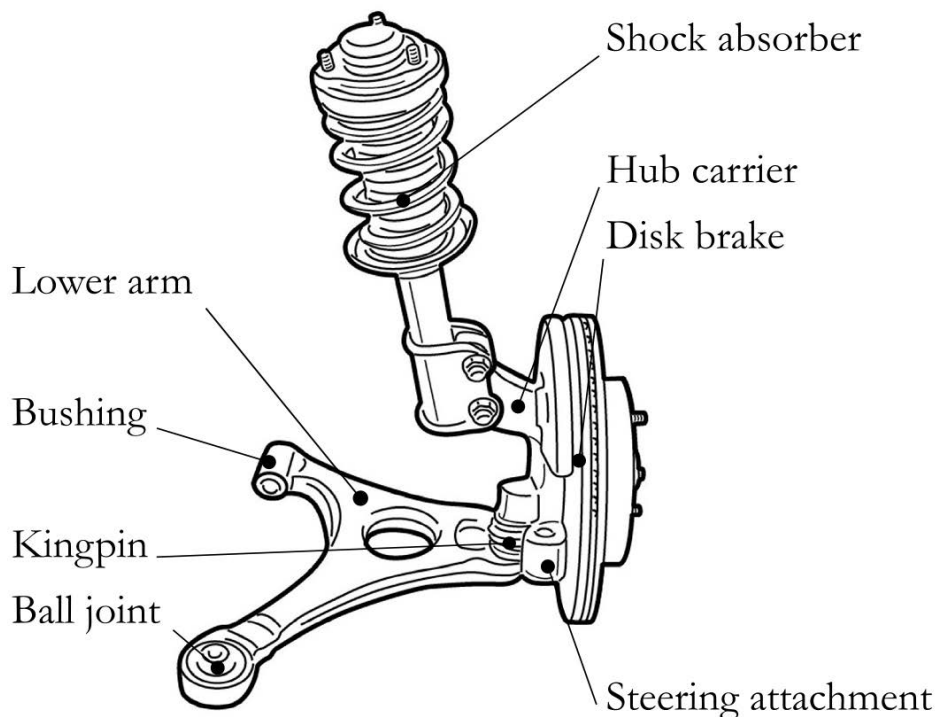


Figure 5.1: Example of MacPherson strut

5.3. Suspension Kinematics and Optimization

The design process begins with the scheme shown in Figure 5.2, which illustrates the main kinematic points of the suspension:

- Point **A** is the connection of the shock absorber to the chassis and represents the upper kingpin.
- Point **B** is the lower kingpin, while **C1** and **C2** are the attachment points between the lower arm and the chassis.
- Points **E** and **D** define the wheel center and the wheel-road contact point, respectively.
- Points **F** and **G** define the position of the steering rod.

The lower arm rotates around the axis defined by **C1** and **C2**, while the spherical joints at **A** and **B** define the steering axis.

The coordinates of these kinematic points are listed in Table 5.1. Some points are fixed according to the vehicle layout, while others can vary within defined limits; these variable points represent the design variables for the optimization problem. Specifically, the optimization aims to determine the values of the five design variables y_B , z_B , y_{C1} , z_{C1} , and y_G in order to meet the target values for camber and steering angles at different suspension extensions, as listed in Table 5.2.

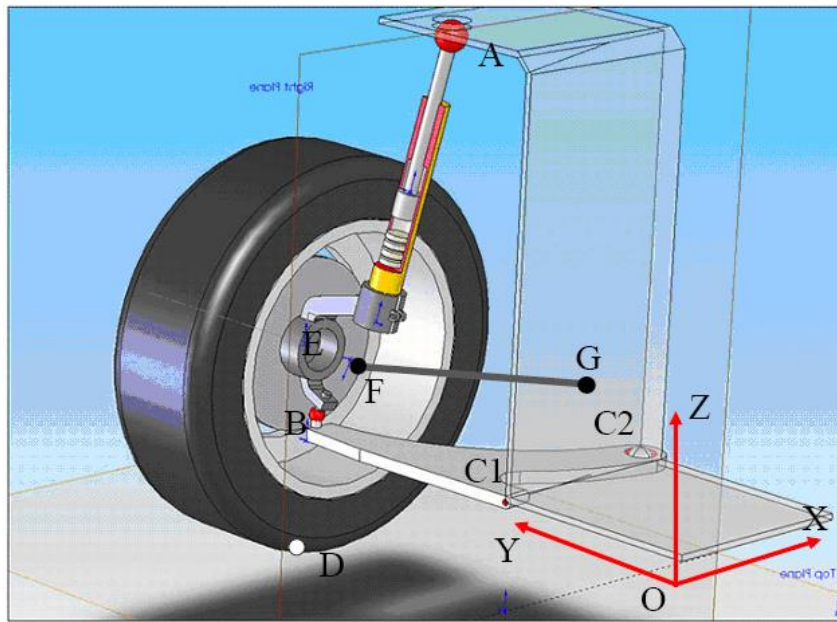


Figure 5.2: Main kinematic points of a MacPherson suspension

Table 5.1: Main kinematic points

Point	Description	x (mm)	y (mm)	z (mm)
A	Rotation centre of the dome	31.8	593	604
B	Ball joint between hub carrier and lower arm	-9.1	[715, 735]	[-70, -50]
C1	Joint between chassis and lower arm	-10	[390, 410]	[-65, -45]
C2	Joint between chassis and lower arm	311	y_{C1}	$z_{C1} + 4.6$
D	Tyre-road contact point	-1.6	773.5	-226.8
E	Wheel center	-1.6	768.4	65.1
F	Steering joint on the hub carrier	136.9	682.7	88.5
G	Steering joint on the chassis (centred steer)	170	[300, 320]	86

Table 5.2: Target values of camber and steer angle

	Nominal position	Full compression (80 mm)	Full extension (95 mm)
Camber angle γ	-1°	$-0^\circ 26' 30''$	$+0^\circ 28' 24''$
Steer angle δ	$-0^\circ 03' 17''$	$-0^\circ 46' 22''$	$+0^\circ 26' 10''$

Since the six target values reported in Table 5.2 require fitting two different curves—the camber angle and the steering angle—the optimization problem is solved using the **Weighted Sum Method**. In this approach, an aggregate objective function is defined as a weighted sum of two individual objective functions, which are calculated as the **root mean square errors (RMSE)** of the camber and steering angles:

$$\phi(x) = \lambda_1 \cdot f_{\text{ob1}} + \lambda_2 \cdot f_{\text{ob2}} = \lambda_1 \cdot \sqrt{\sum_{i=1}^{n_c} (\gamma_i - \gamma_i^0)^2} + \lambda_2 \cdot \sqrt{\sum_{i=1}^{n_c} (\delta_i - \delta_i^0)^2} \quad (5.1)$$

where:

- γ_i and δ_i are the computed values of the camber and steering angles at the fitting points;
- γ_i^0 and δ_i^0 are the target values reported in Table 5.2;
- n_c is the number of objectives in the fitting procedure ($n_c = 3$ in this case);
- λ_1 and λ_2 are the weighting coefficients for the camber and steering angle objectives, respectively.

The values of γ_i and δ_i are computed using a **Matlab** function based on the **Matlab SimMechanics** package (Figure 5.3). This function takes as input the three-dimensional coordinates of the kinematic points previously described in Table 5.1 and returns the corresponding values of the characteristic angles.

The fitting problem is then solved using the Matlab built-in optimization function **fmincon**. The weighting coefficients are varied between 0 and 1 in order to obtain different solutions belonging to the **Pareto-optimal set**.

Figure 5.4 shows the solution obtained for $\lambda_1 = \lambda_2 = 0.5$, for which the camber and steering curves closely match the target values, with a maximum deviation error of 0.28%. The coordinates of the kinematic points corresponding to the fitted curves are reported in Table 5.3.

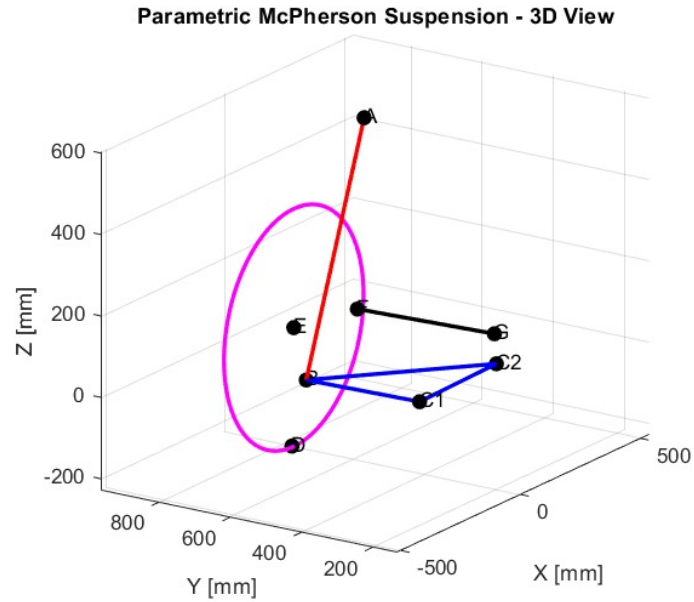


Figure 5.3: Matlab SimMechanics model of the suspension

Table 5.3: Optimal value of the kinematic points by curve fitting

Design variable	Minimum (mm)	Maximum (mm)	Optimized (mm)
y_B	715	735	727.84
z_B	-70	-50	-53.77
y_{C1}	390	410	408.69
z_{C1}	-65	-45	-47.82
y_G	300	320	319.52

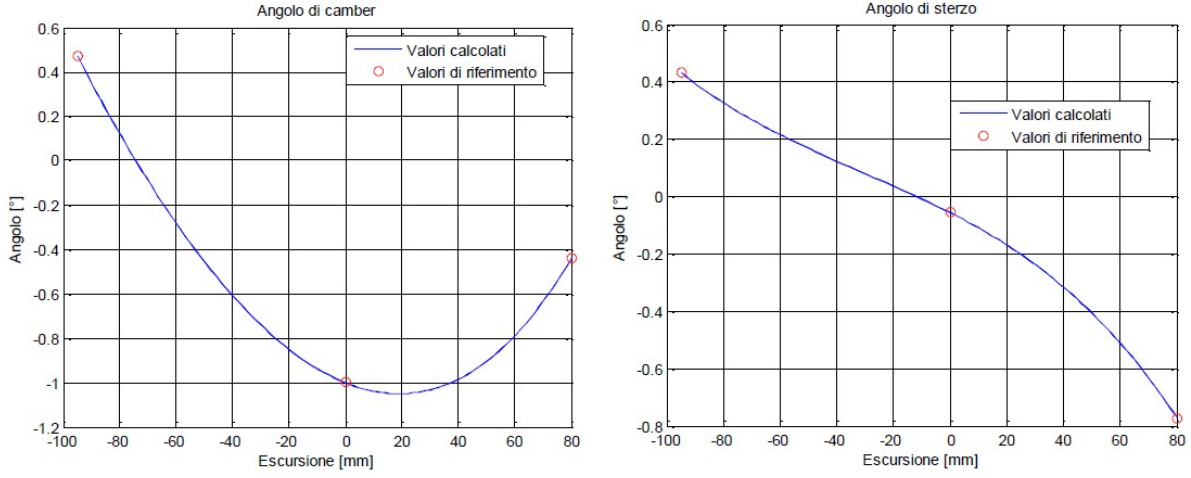


Figure 5.4: Camber (left) and steering (right) variation as function of the stroke after kinematic optimization. The blue line is the output computed by the SimMechanics model, the red circles are the target values.

5.4. Lower arm forces computation

In order to perform a preliminary design of the lower arm, it is necessary to evaluate the main forces acting on the suspension and to determine how these loads are transmitted to the component. All forces are expressed with respect to the reference frame shown in Figure 5.2.

To account for the most critical loading condition, the simultaneous application of vertical, longitudinal, and lateral forces is considered. Furthermore, the vertical load F_z is increased by means of a safety factor $k = 2.5$ in order to account for road irregularities and bump effects:

$$Z = kF_z \quad (5.2)$$

Since commonly adopted models for the computation of longitudinal and lateral tire slip are excessively complex for the purposes of this analysis, a simplified approach is adopted. The longitudinal force F_x and the lateral force F_y are assumed to be proportional to the vertical load:

$$X = \mu_x Z \quad Y = \mu_y Z \quad (5.3)$$

where $\mu_x = 1.2$ and $\mu_y = 0.7$ are appropriate friction coefficients.

At this stage, once the forces acting at the spherical joint are known, the reaction forces at the lower arm bushings can be evaluated. The lower arm is connected to the chassis through two bushings, which are assumed to transmit loads along specific directions defined by their geometric orientation.

The reaction forces are computed by enforcing static equilibrium in two orthogonal planes. The equilibrium is first evaluated in the **Y–Z plane**, where the lateral force F_y and the vertical force F_z generate moments about the spherical joint. These moments are balanced by the bushing reaction force B_u , whose direction is defined by the inclination angle α :

$$B_u = \frac{F_y |D_z - A_z| - F_z |D_y - A_y|}{\cos \alpha |B_z - A_z| - \sin \alpha |B_y - A_y|} \quad (5.4)$$

Similarly, the equilibrium in the **X–Z plane** is considered to evaluate the effects of the longitudinal force F_x and the vertical force F_z . The resulting bushing reaction force B_v acts along a direction defined by the inclination angle γ :

$$B_v = \frac{F_x |D_z - A_z| - F_z |D_x - A_x|}{\cos \gamma |B_z - A_z| - \sin \gamma |B_x - A_x|} \quad (5.5)$$

The angles $\alpha = 1.9^\circ$ and $\gamma = 1.88^\circ$ describe the orientation of the bushing reaction directions.

By applying the equilibrium equations described above, the reaction forces at the lower arm bushings can be computed. For the considered loading condition, the resulting reaction forces are:

$$B_u = 5.88 \times 10^3 \text{ N} \quad (5.6)$$

$$B_v = 1.40 \times 10^4 \text{ N} \quad (5.7)$$

The results show that the reaction force associated with the longitudinal loading condition (B_v) is significantly higher than the one related to the lateral loading condition (B_u). This behavior is mainly due to the higher magnitude of the longitudinal force and the corresponding lever arms involved in the equilibrium of the system.

These reaction forces represent the loads transmitted from the suspension to the chassis through the lower arm bushings and are therefore used as input for the subsequent structural and finite element analysis of the lower arm.

5.5. Material Selection

As a starting point for defining the geometry of the component, the suspension lower arm is assumed to be manufactured from **6061-T6 aluminium alloy** using a forging process.

Aluminium 6061 is a medium-strength, heat-treatable alloy belonging to the 6000 series, characterized by good corrosion resistance, high toughness, and excellent manufacturability. Due to its balanced combination of mechanical strength, formability, and fatigue resistance, 6061-T6 is widely used in automotive structural components, including suspension systems, as well as in aerospace and general engineering applications. Its good forgeability and recyclability make it particularly suitable for mass-produced forged components subjected to moderate to high loads.

The conventional manufacturing cycle begins with the casting of 6061 aluminium billets. These billets are subsequently shaped through a closed-die forging process, in which a hammer and an anvil fully enclose the workpiece, forcing the material to flow and fill the die cavity. The excess material, known as flash, is then removed using a trimming die. Finally, a sequence of machining operations is carried out to obtain the functional surfaces and holes, resulting in the final component.

The main mechanical properties of the 6061-T6 aluminium alloy, adopted for the structural analysis, are summarized in Table 5.4.

Table 5.4: Mechanical properties of 6061-T6 aluminium alloy

Property	Value
Density	2.7 g/cm ³
Yield strength	275 MPa
Tensile strength	310 MPa
Elastic modulus	69 GPa
Poisson's ratio	0.33

The shape of the lower arm has to take in consideration the configurations of a steering angle of 37° inward and of 32° outward the nominal position.

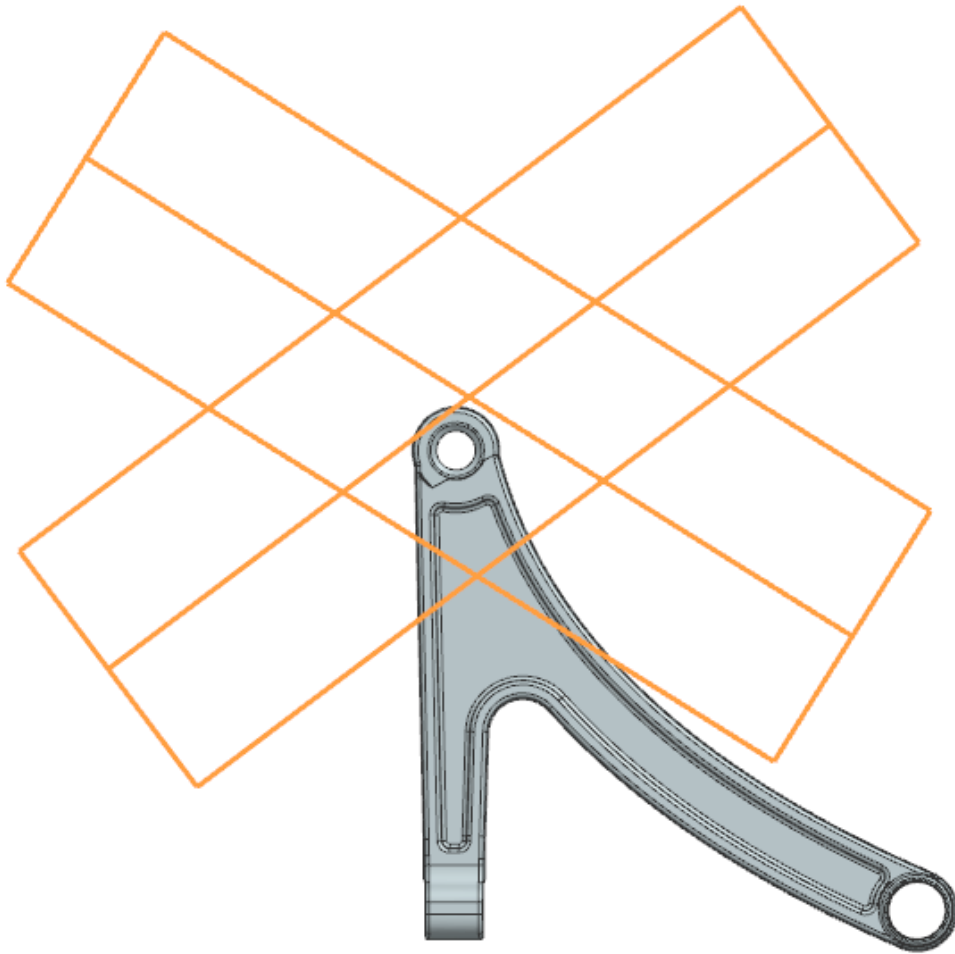


Figure 5.5: Steerign angles of the wheel

5.6. Finite Element Analysis of the Lower Arm

The finite element analysis (FEA) is performed using the **SolidWorks Simulation** environment in order to evaluate the displacement and stress distributions within the component. Due to the geometric symmetry of the lower arm, only half of the body is considered in the simulations, thus reducing the computational cost.

The analysis is carried out by applying the following boundary conditions and loads:

- **Spherical joint constraint** applied at point C2;
- **Pin constraint** applied at point C1, allowing translation along the pin axis and free rotations;
- **Roller constraint** applied at point B, preventing vertical translation.

The reaction forces computed from the static equilibrium analysis are applied as external loads, with magnitudes:

$$B_u = 5.88 \times 10^3 \text{ N}, \quad B_v = 1.40 \times 10^4 \text{ N}. \quad (5.8)$$

Boundary conditions and loads are applied on the cylindrical surfaces corresponding to the attachment regions of the lower arm.

The stress distribution and displacement of the final component are shown in Figures 5.6 and 5.7. As can be observed, the FEM simulation predicts a maximum von Mises stress of 136 MPa and a maximum displacement of 1.59 mm.

Further details of the analysis are summarized in Table 5.5.

Table 5.5: Summary of FEA results

Parameter	Value
Maximum von Mises stress	136 MPa
Maximum displacement	1.59 mm
Yielding safety factor	2.01
Mass	1.507 kg

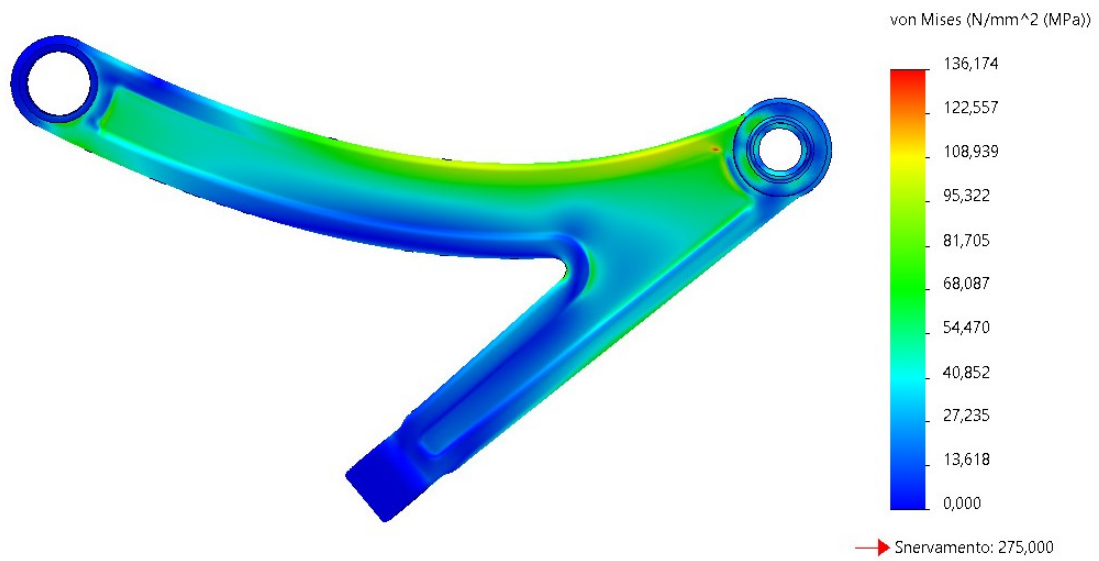


Figure 5.6: Stress Distribution of the Lower Arm

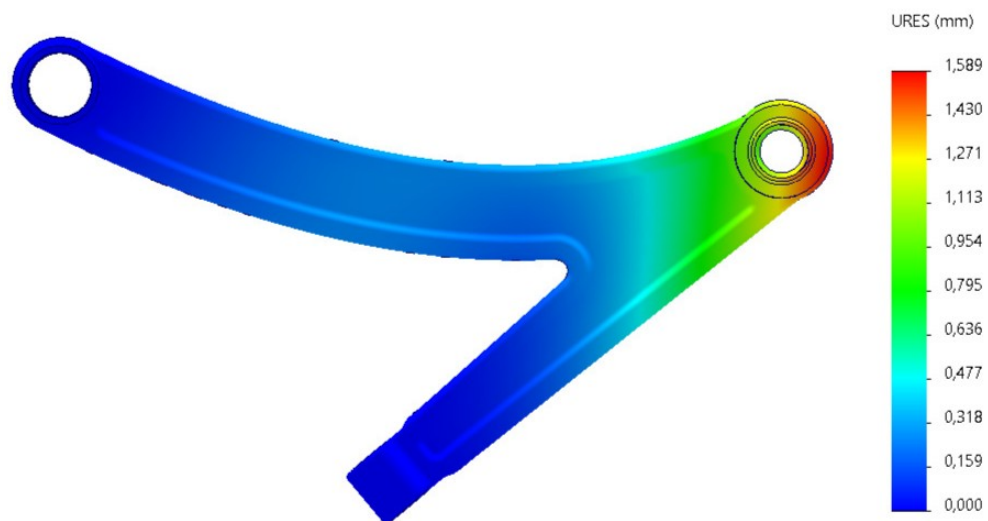


Figure 5.7: Displacements of the Lower Arm

5.7. Suspension Assembly

The final stage of the design process involved the development of a complete suspension assembly, incorporating all the essential components while ensuring proper integration and overall functionality. The selected suspension layout is the **MacPherson strut**, which is widely adopted in modern vehicles due to its simple architecture, cost-effectiveness, and efficient use of the available space.

Based on the previously obtained results and calculations, the suspension assembly was modelled using **SIEMENS NX**. All components were positioned according to the defined kinematic points and geometric constraints, resulting in a coherent and functional assembly. The final suspension layout is shown in the figure below.

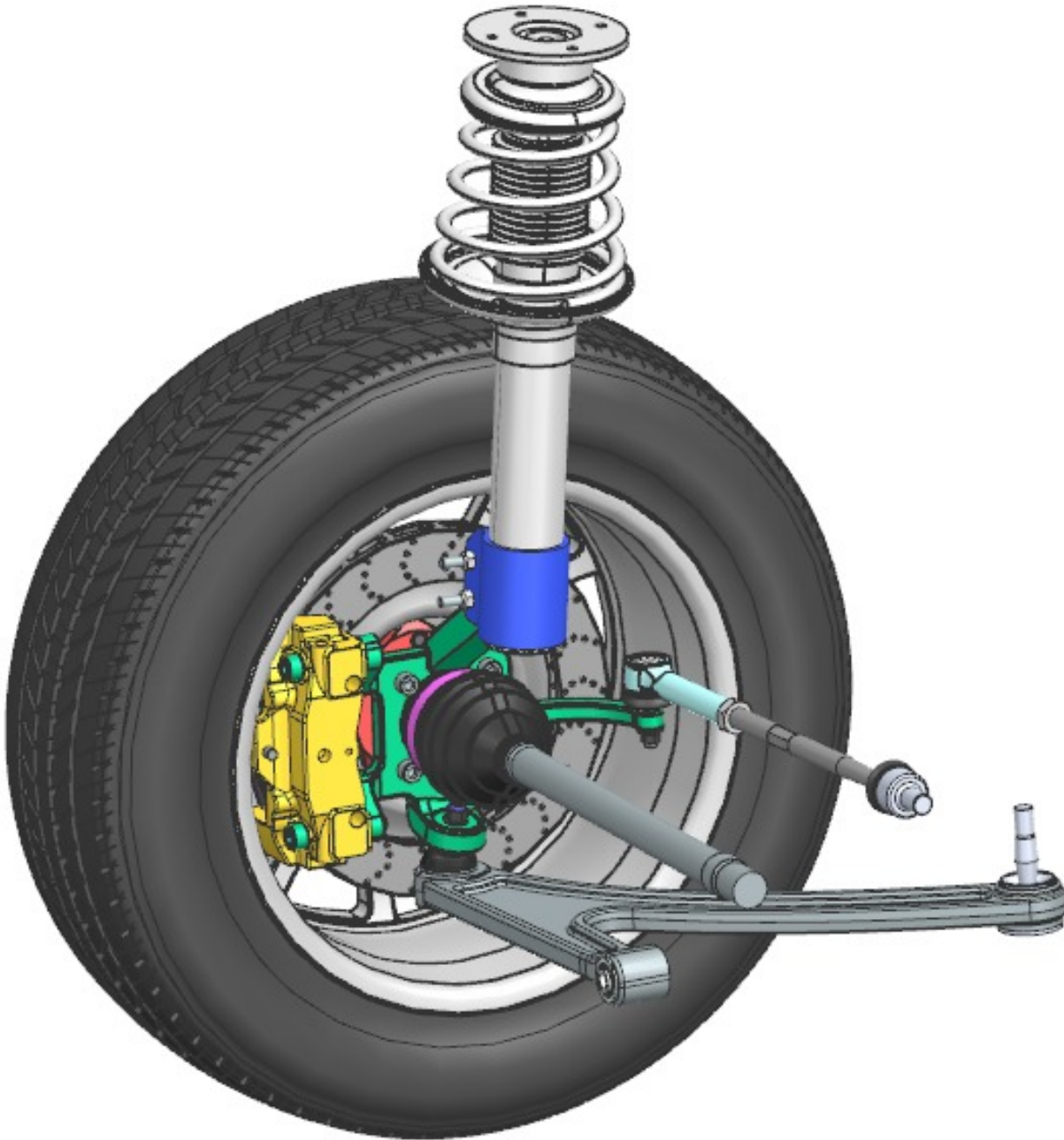


Figure 5.8: Suspension Assembly

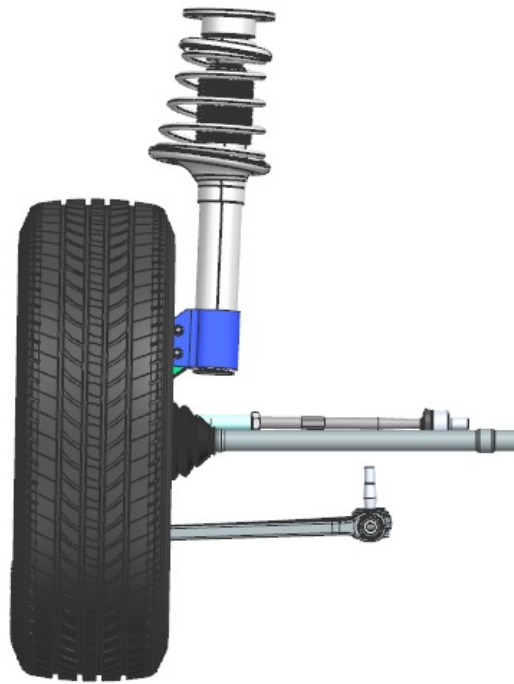
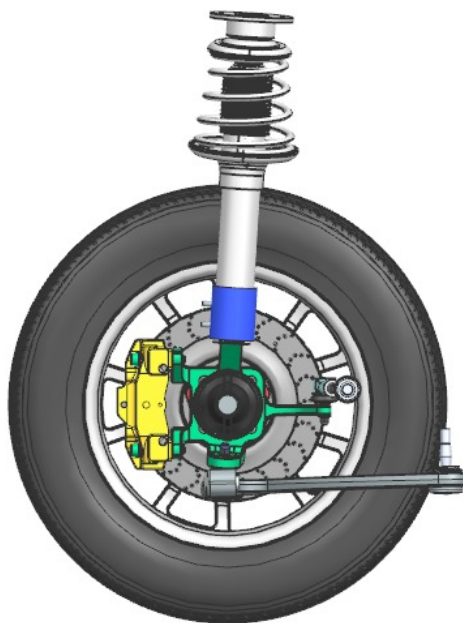
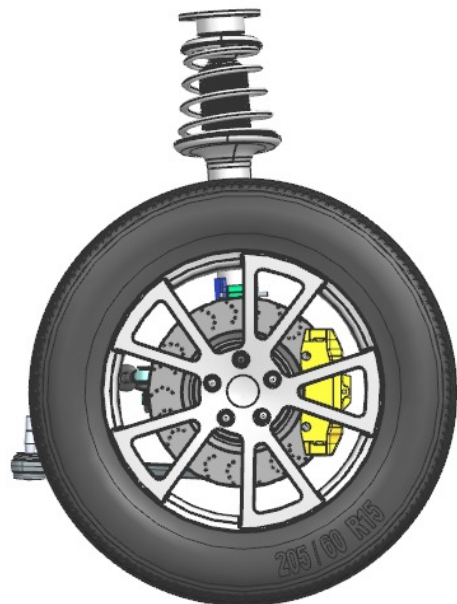


Figure 5.9: Suspension Assembly - Front View



(a) Suspension Assembly - Internal View



(b) Suspension Assembly - External View

6 | Design of Aluminum Wheels

6.1. Introduction and Specifications

The design and assessment of lightweight aluminium wheels for the Porsche 718 I (2020) emphasize a balance between safety, performance, and style. This analysis utilizes digital twins and finite element modeling to simulate dynamic cornering fatigue tests based on ISO and SAE standards (ISO 3006, SAE J328, SAE J2562).

The standard tire specification for the vehicle is **245/35 ZR20 95Y**. Key parameters derived for the analysis include:

- **Load Index (95):** Corresponds to a maximum load capacity of 690 kg per tire.
- **Speed Rating (Y):** Corresponds to a maximum safe operational speed of 300 km/h.

6.2. Bending Moment Calculation

To evaluate wheel durability during cornering, the bending moment (M) is established using parameters such as friction coefficient, tire radius, and vertical static load. The bending moment acting on the wheel is defined as:

$$M = (\mu R + d) \cdot F_v \cdot S \quad (6.1)$$

Where:

- M : Bending moment [Nm]
- μ : Assumed friction coefficient between tyre and road (0.7)
- R : Tyre radius [m]
- d : Inset or outset of the wheel [m]
- F_v : Maximum vertical static load or tire load rating [N]

- S : Accelerated test factor ($S=20$)

6.3. Finite Element Modeling (FEM)

The dynamic cornering fatigue test is simulated using a linear elastic finite element model with the following material properties:

- **Elastic Modulus (E):** 70 GPa
- **Poisson's Ratio (ν):** 0.3
- **Element Type:** Quadratic tetrahedral elements (8 mm typical, refined to 4 mm in high-stress regions).

6.4. Dynamic Stress Computation

To approximate the complex stress state under a rotating moment, the stress states from two orthogonal unitary bending moments are superposed. The dynamic stress $\sigma(\alpha)$ is computed as a function of the rotation angle α :

$$\sigma(\alpha) = M(\sigma_{Mz} \cos \alpha + \sigma_{Mx} \sin \alpha) \quad (6.2)$$

6.5. Fatigue Life Estimation

6.5.1. Sines Fatigue Criterion

The Sines criterion is employed to calculate the equivalent stress amplitude (σ_S) from multiaxial stress components:

$$\sigma_S = \sqrt{\sigma_{x,a}^2 + \sigma_{y,a}^2 + \sigma_{z,a}^2 - (\sigma_{x,a}\sigma_{y,a} + \sigma_{x,a}\sigma_{z,a} + \sigma_{y,a}\sigma_{z,a}) + 3(\tau_{xy,a}^2 + \tau_{xz,a}^2 + \tau_{yz,a}^2)} \quad (6.3)$$

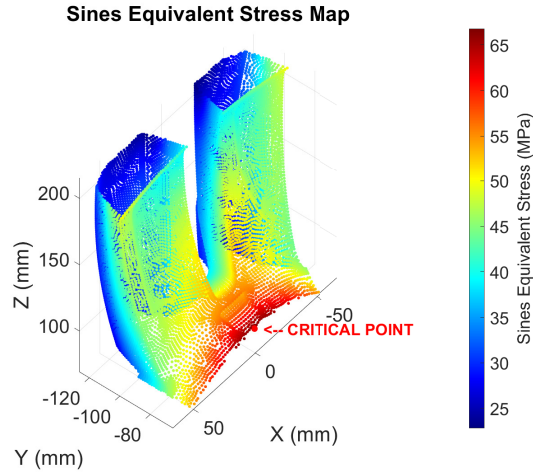


Figure 6.1: Sines stresses distribution on the rim.

6.5.2. Wohler Curve and Cycle Prediction

The fatigue life (N_f) is predicted using the logarithmic representation of the Wohler curve. The relationship between the stress limit (σ_{lim}) and cycles to failure is expressed as:

$$\log(\sigma_{lim}) = \log(\sigma'_f) + b \log(2N_f) \quad (6.4)$$

Where:

- σ'_f : Fatigue strength coefficient.
- b : Slope of the Wohler curve characterizing the material's fatigue behavior.

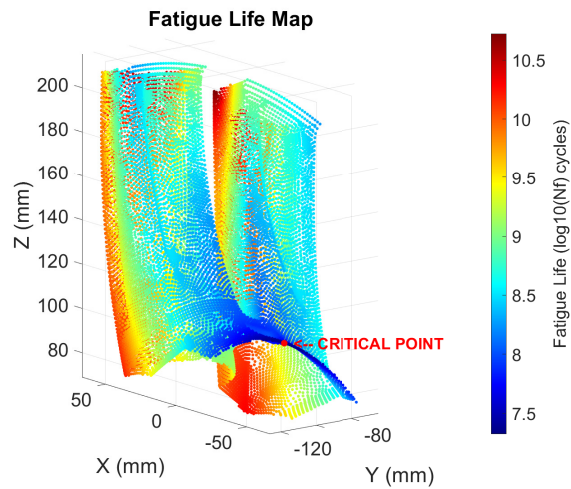


Figure 6.2: Number of admissible cycles per point.

The fatigue assessment of the wheel illustrated through Sines equivalent stress and fatigue life maps, identifies critical stress concentrations and potential failure points. High Sines equivalent stress regions, marked in red, are concentrated at the close to the inner mating surface of the wheel, corresponding to areas with the shortest fatigue life. These high-stress zones are most susceptible to fatigue failure under cyclic loading. Conversely, the outer portion of the wheel, exhibiting lower stress levels and longer fatigue life, is less prone to failure. These insights highlight the necessity for targeted design enhancements or reinforcements at the outer edges and inner corners to improve the wheel's overall durability and service life.

These high-stress zones are most susceptible to fatigue failure under cyclic loading, aligning with the ISO 3006 standard for road vehicle wheels and the dynamic cornering fatigue test methodology, which evaluates the wheel's performance under simulated bending moments and stress states. Where the minimum number of fatigue cycles the wheel has to sustain to pass the test is 50,000 for an Accelerated test factor (S) of 2 for Aluminium and our rim at the weakest point sustains for about 10^7 cycles.

The stress computation involved simulating two orthogonal unitary bending moments combined and scaled to represent the rotating bending moment. The central portion of the wheel, displaying lower stress levels and longer fatigue life, is less prone to failure, indicating effective load distribution and structural integrity in this area.

These findings underscore the necessity for targeted design enhancements or reinforcements at the outer edges and inner corners to improve the wheel's overall durability and service life. By addressing these critical stress zones, it is possible to enhance the wheel's performance, ensuring it meets safety standards and performs reliably under real-world conditions.

6.6. Conclusion

This comprehensive methodology, executed via MATLAB for stress tensor analysis and FEM for structural simulation, ensures the reliability of aluminium wheels under dynamic loads. By accounting for multiaxial stress distributions and material-specific fatigue curves, the analysis contributes significantly to the overall safety and efficiency of the Porsche 718 I.

7 | Seminars Summary

7.1. Visit at Drismi

7.1.1. Laboratory Objective

The primary objective of this laboratory session was to physically evaluate and comprehend the disturbances experienced by a driver when a vehicle is subjected to excitations at its resonance frequencies. The study focuses on two critical frequencies:

- $f = 1 \text{ Hz}$: The natural frequency of the sprung mass.
- $f = 15 \text{ Hz}$: The natural frequency of the unsprung mass.

7.1.2. Laboratory Procedure

Using a vehicle simulator, the subjective differences between 1 Hz and 15 Hz excitations were analyzed. Following the isolated frequency tests, a simulation was conducted on a road surface characterized by repeated cleats to observe the combined effects of road irregularities on vehicle dynamics and driver comfort.

7.1.3. Subjective Frequency Response Function (FRF)

During the simulation, the driver acted as a qualitative human FRF estimator, recording the physical sensations associated with specific frequency inputs.

$f = 1 \text{ Hz}$ (Sprung Mass Resonance)

At 1 Hz, the vehicle exhibited high-amplitude vertical oscillations, with motion alternating between the front and rear axles.

- **Physiological Impact:** The motion caused a heavy sensation of internal organ displacement.
- **Prolonged Effect:** Extended exposure resulted in head aching.

$f = 15 \text{ Hz}$ (Unsprung Mass Resonance)

At 15 Hz, the displacement amplitudes were significantly lower; however, the high-frequency content shifted the disturbance to the extremities.

- **Physiological Impact:** Intense vibration was felt through the footplate.
- **Post-Simulation Effect:** Muscular rigidity in the plant of the feet was reported after exiting the cockpit.

Road Cleats Evaluation

When traversing the runway with repeated cleats, the subjective feeling was dominated by the first resonance (1 Hz) rather than the second (15 Hz). The cleats primarily induced a high-amplitude pitching motion of the vehicle body.

7.1.4. Acceleration Data

Preliminary quantitative data was acquired using a mobile device accelerometer (excluding gravity (g)) to correlate subjective feelings with physical acceleration peaks.

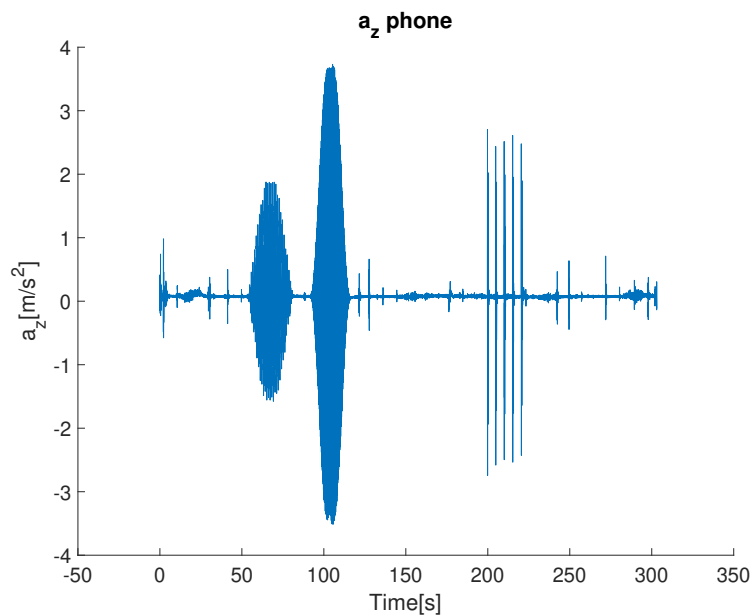


Figure 7.1: Vertical accelerations experienced during test..

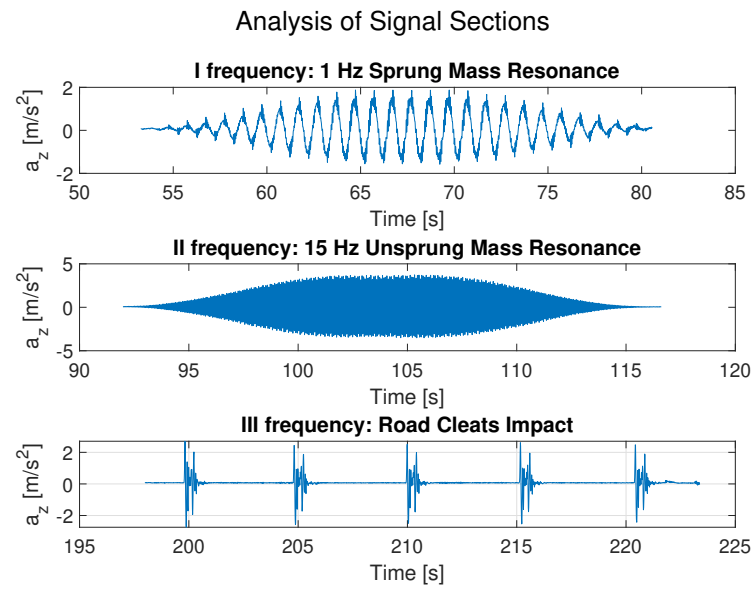


Figure 7.2: Truncated signal, with tests at different excitation frequencies.

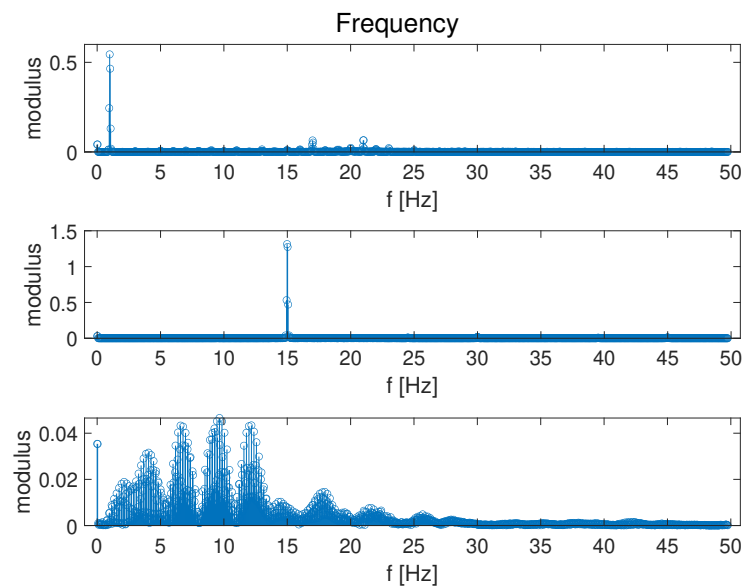


Figure 7.3: Spectrum of the signal for each signal section.

The signal shows the two different tests at different imposed frequencies, the tests had also a difference in amplitude and the

7.1.5. Conclusions and Next Steps

The simulation successfully demonstrated that prolonged frequency excitation, particularly at high frequencies, results in physiological disturbances for the pilot. The lab is helpful to understand the importance of the suspension system for driver health and comfort. Future analysis will involve the post-processing of the accelerometer data to calculate the objective FRF and compare it with the subjective observations recorded.

7.2. Visit at LaST

The LaST (Laboratorio Sicurezza Trasporti) at Politecnico di Milano is dedicated to advancing the safety of various transportation systems. Researchers at LaST focus on minimizing injury risks for occupants of vehicles such as cars, trains, and helicopters. Their work encompasses both passive and active safety measures, aiming to improve overall transport safety through innovative research and development. We were given the opportunity to visit LaST and see in person various examples of the work that is being carried out. Various types of suspensions were exposed. An Alfa Romeo Giulia was lifted to show its refined systems: a double-wishbone with virtual joint of the lower arm at the front and a very sophisticated multi-link at the rear, with multiple-stage vibration filtering. It was possible to see a more conventional layout on a jeep wrangler, with Mac-Pherson at the front and torsion beam at the rear. Finally, the rear axle assembly of an Acura was on display: it was possible to see clearly all the features of its multi-link suspension. In addition to this, several components were on view: suspension arms, synchronizer, differentials, hubs, hub carriers among others. This allowed to study in detail their function in the vehicle and their possible manufacturing processes. Among LaST equipment it was also possible to see the InTenso test rig: a carrying system (steel cage), which bears the body under test and measures the inertia properties of the studied vehicle (inertia tensor and centre of gravity position) using load cells and encoders. Another remarkable instrument is Ruotavia: a horizontal steel drum apparatus useful to design and test road tyres. Lastly, the lab offers another way to measure and assess tyre-road contact forces: a purpose-designed wheel rim equipped with many extensometers can measure the forces going through the wheel. The visit to the LaST laboratory was valuable, providing a hands-on experience with various components and an opportunity to observe their functionality. It was particularly interesting to see the advanced tools and equipment used for design and development in the automotive sector. This firsthand exposure to real components and innovative research methods significantly enhanced our understanding of the complexities involved in ensuring transportation safety.

7.3. Cromodora Aluminum Alloy Wheels

7.3.1. Introduction

The guest lecture focused on "Aluminum Alloy Wheels at Cromodora Wheels," presented by C. Sorlini. The session provided a comprehensive overview of the company's operations, the various types of aluminum alloy wheels, manufacturing processes, quality planning frameworks, and ongoing research initiatives.

7.3.2. Company Overview

Cromodora Wheels has a prestigious history dating back to 1928. Key milestones include:

- **1966:** Production of the first high-pressure magnesium wheel.
- **1969:** Commencement of aluminum alloy wheel production.

Currently, the company operates two primary state-of-the-art plants:

- **Ghedi, Italy (60,000 sqm):** Specializes in wheels up to 23 inches, exotic finishes, and advanced machining.
- **Mosnov, Czech Republic (96,000 sqm):** Focused on high production runners and automation, with a nominal capacity of 2 million units.

7.3.3. Aluminum Alloy Wheel Types

The lecture distinguished between two primary manufacturing categories for alloy wheels:

- **Cast Wheels:** The most common type, produced via Low-Pressure Die Casting (LPDC) or Gravity Die Casting (GDC). LPDC fills the die from the bottom, while GDC utilizes gravity. These methods provide a balance of design variety and cost-effectiveness.
- **Forged Wheels:** Offer superior material characteristics and lower weight. They are categorized as:
 - *Net Forging:* Cost-effective but limited in design variety.
 - *Blank Forging:* Allows for intricate, high-performance designs through subsequent machining.

7.3.4. Technical Features and Specifications

The design and fitment of wheels depend on several critical characteristics:

- **Geometrical Characteristics:** Style, diameter, rim width, Pattern Cycle Diameter (PCD), offset, hump, and rim type.
- **Interfaces:** Tire compatibility, brake caliper packaging profiles, hub diameter, center cap design, and valve placement.
- **Cosmetics and Dynamics:** Finishes range from fully painted to diamond-turned. Dynamic properties include lateral/vertical load capabilities, natural frequencies, and stiffness.
- **Functional Specifications:** Includes mechanical, metallurgical, and corrosion resistance testing, alongside strict limits on dynamic unbalance.

7.3.5. Advanced Product Quality Planning (APQP)

Cromodora utilizes the APQP framework to manage product quality across five phases:

1. **Bid/Win:** Feasibility studies, customer pitching, and business case definition.
2. **Product Design:** Design, dimensioning, and the final design freeze.
3. **Product Tooling and Validation:** Tooling creation, sampling, and engineering approval.
4. **Product Approval for Production:** Preseries production and customer approval.
5. **Production Ramp-Up:** Full capacity installation and project handover to the production plant.

7.3.6. Learning Outcomes

The lecture provided several key technical insights:

- **Manufacturing:** Appreciation for the trade-offs between LPDC (complex designs/low cost) and forging (weight reduction/performance).
- **Quality Assurance:** Understanding the rigorous standards required for automotive durability.
- **Innovation:** Exposure to cutting-edge technologies like *flow forming* and advanced R&D initiatives.

7.3.7. Conclusion

The guest lecture highlighted the complexities of wheel manufacturing and the importance of stringent quality control. Cromodora Wheels' commitment to innovation and robust engineering processes provides a vital roadmap for producing high-performance components in the modern automotive industry.

7.4. Guest lectures by Capgemini

7.4.1. Introduction

Capgemini conducted a comprehensive guest lecture focusing on two critical aspects of engineering: Functional Safety lectured by Elena Caovilla and Design Failure Modes and Effects Analysis (FMEA) lectured by Elisa Compierchio. The lecture was designed to provide in-depth knowledge and practical insights into these essential topics, widely applicable in various industries, particularly automotive and aerospace. The following sections provide a detailed summary of the teachings and learnings from the lecture, divided into two main parts: Functional Safety and FMEA, each followed by reflections on the learnings and outcomes.

7.4.2. Functional Safety - Capgemini

Functional Safety is defined as the part of the overall safety relating to the equipment under control (EUC) and its associated control system that depends on the correct functioning of the safety-related system. It is achieved when specified safety functions satisfy their required safety performance. While zero risk is not possible, safety is achieved by reducing risk to a tolerable level.

7.4.3. Key Learning Modules

- **Active vs. Passive Safety:** Distinguished between passive (reducing accident effects, e.g., airbags) and active safety systems (preventing accidents, e.g., ABS).
- **Safety Standards:** Analyzed the **IEC 61508** series as a basic functional safety standard for the E/E/PE lifecycle. Studied **ISO 26262** for road vehicles, which addresses malfunctioning behavior in automotive electrical/electronic systems.
- **HARA Methodology:** Developed Hazard Analysis and Risk Assessments to specify safety goals and ASILs by evaluating three parameters:

- **Severity (S):** Extent of harm to individuals (S0–S3).
- **Exposure (E):** Probability of operational situations that can be hazardous (E1–E4).
- **Controllability (C):** Ability to avoid harm through timely reactions of the persons involved (C0–C3).
- **ASIL Classification:** Learned risk classification (ASIL A–D) where ASIL D represents the highest degree of hazard (e.g., airbags, steering).

7.4.4. Safety of the Intended Functionality (SOTIF)

Beyond traditional functional safety, the course addressed **SOTIF (ISO/PAS 21448)**, which is defined as the absence of unreasonable risk due to hazards resulting from functional insufficiencies of the intended functionality or by reasonably foreseeable misuse by persons.

Distinction from ISO 26262: While ISO 26262 addresses malfunctioning behavior caused by system faults (systematic and random), SOTIF focuses on systems that may fail to operate safely even without a hardware or software failure. **Interdependence:** To achieve SOTIF, it is generally assumed that the item has already been developed and produced in accordance with ISO 26262. **The Safety Triangle:** SOTIF works in conjunction with **Functional Safety (FuSa)** and **Cybersecurity (ISO/SAE 21434)** to ensure the overall safety of the item.

7.4.5. Practical Application: Active Suspension System

Conducted a guided exercise on an **Active Suspension System** to derive Safety Goals and Safe States:

- **BCM_SG1:** Avoid Fire and thermal runaway on the MVB (ASIL C).
- **BCM_SG2:** Avoid Human Electrocution (48V BUS discharged at Voltage < 60V) (ASIL B).
- **BCM_SG3:** Avoid fire on 48V cables (ASIL A).

7.4.6. FMEA-FMECA Capgemini

FMEA is a systematic preventive analysis tool used within the **Preventive Reliability** approach to achieve expected quality and aim for "zero defects" throughout the product lifecycle. The methodology is applied during design (**D-FMEA**) to prevent issues and during production through problem-solving to investigate root causes of occurred failures.

Methodology and Types

- **Design FMEA (D-FMEA):** Focuses on identifying potential failure modes of a new design item, their effects on the user, and related potential root causes.
- **Process FMEA (P-FMEA):** Utilizes critical design characteristics identified in the D-FMEA as inputs to examine manufacturing and assembly processes.
- **7-Step Approach:** The course utilized the **AIAG & VDA FMEA Handbook** 7-step process: (1) Planning & Preparation, (2) Structure Analysis, (3) Function Analysis, (4) Failure Analysis, (5) Risk Analysis, (6) Optimization, and (7) Results Documentation.

7.4.7. Risk Assessment and Outcomes

Risk is prioritized using the **Risk Priority Number (RPN)**, calculated as:

$$\text{RiskPriorityNumber} = \text{Severity} \times \text{Occurrence} \times \text{Detection} \quad (7.1)$$

The course defined intervention thresholds (e.g., $\text{RPN} > 100$ requiring immediate improvement) and an Action Priority scale (High, Medium, Low) based on the Severity-Occurrence-Detection matrix.

7.4.8. Case Study: Front Bumper and ADAS Radar

The practical application focused on a D-FMEA for a Front Bumper for the Stralis MY24. A key failure analyzed was **ADAS malfunctioning** (Severity 9) due to **interference between the foldable step and the radar view**. The optimization involved stack-up analysis to ensure installation met supplier prescriptions, resulting in a reduced RPN from 126 to 72.

7.4.9. Functional Safety Framework

The lecture provided a framework for understanding safety measures through:

- **HARA Methodology:** Identifying hazards based on **Severity (S)**, **Exposure (E)**, and **Controllability (C)** to derive **Safety Goals** and **ASIL** ratings.
- **E/E/PE Lifecycle:** Understanding the lifecycle from concept through decommissioning, distinguishing between **Passive Safety** (reducing accident effects like airbags) and **Active Safety** (preventing accidents like ABS).

7.5. Visit at Vizzola Pirelli test track

7.5.1. Introduction

The visit to the Pirelli Proving Ground in Vizzola Ticino provided a fundamental opportunity to observe the practical application of vehicle dynamics theories discussed during the course. The activities focused on the interaction between the tire and the road surface, specifically exploring the limits of adhesion and the transition between different handling behaviors (understeer and oversteer). The experience was divided into two main experimental sessions: the Steering Pad test and a dynamic Handling test on a wet track.

7.5.2. Steering Pad Test

The first session took place on the steering pad, a large circular area with a constant radius. The goal was to characterize the vehicle's steady-state cornering behavior and observe how longitudinal forces affect lateral stability.

Theoretical Framework

As studied in the handling diagram ($\delta - L/R$ vs. a_y/g), a vehicle's behavior is defined by the relative slopes of the front and rear axle characteristic curves. Under nominal conditions at constant speed, the vehicle demonstrated a typical **understeering** behavior, where the steering angle required increases more than linearly with lateral acceleration.

Experimental Observations

During the test, the driver demonstrated the effect of increasing the traction force while maintaining the turn.

- **Traction Influence:** By accelerating mid-corner, the longitudinal force F_x at the rear wheels increases. According to the friction ellipse (Kamm's circle) concept, an increase in longitudinal force reduces the maximum available lateral force F_y .
- **Behavioral Shift:** This reduction in lateral capacity effectively "lowers" the characteristic curve of the rear axle. We observed that the vehicle transitioned from understeering to **oversteering**, requiring the driver to reduce the steering angle or apply counter-steer to maintain the trajectory.

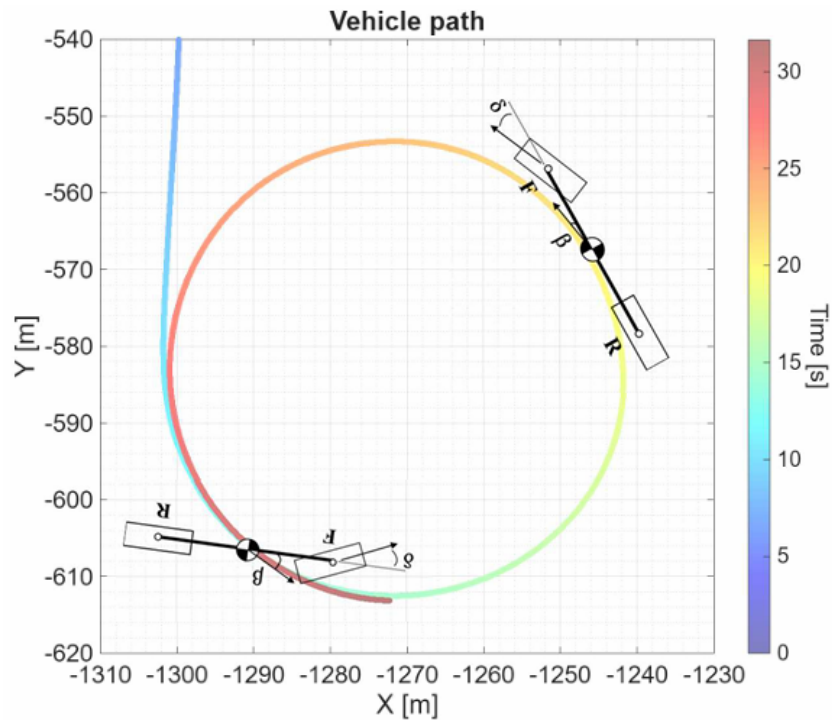


Figure 7.4: Power on manouver

7.5.3. Wet Handling Test Drive

The second part of the visit involved a dynamic test drive on the wet handling circuit. This track is characterized by low friction coefficients (μ), which significantly amplify the vehicle's sensitivity to driver inputs.

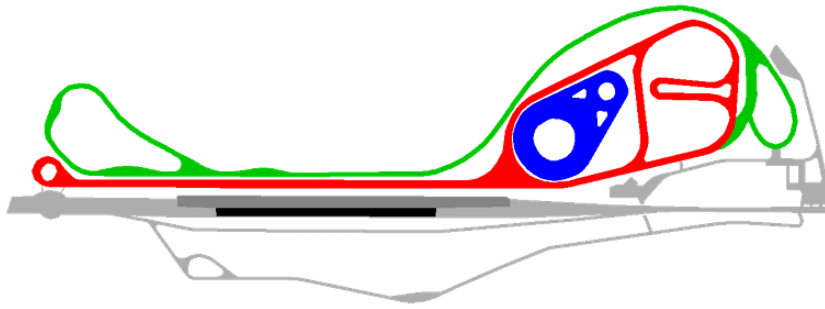


Figure 7.5: Pirelli test track map.

7.5.4. Dynamics in Low Adhesion

Accompanied by a professional Pirelli test driver, we analyzed the vehicle's response to combined maneuvers (braking/accelerating while cornering). In wet conditions, the "safety margin" provided by the friction circle is drastically reduced.

Key Findings

- **Load Transfer and Stability:** The test highlighted how vertical load transfer during braking (dive) and acceleration (squat) changes the grip balance between the axles.
- **Power Oversteer:** On the wet surface, it was easier to reach the adherence limit of the rear axle. Even moderate throttle inputs resulted in significant slip angles at the rear, demonstrating how a rear-wheel-drive (RWD) vehicle manages power-on oversteer in low-grip scenarios.
- **Driver Feedback:** The test driver emphasized the importance of tire transient response—how quickly the tire regains grip after a slide—which is a critical parameter for Pirelli when developing high-performance tires.

7.5.5. Conclusions

The visit to Vizzola was instrumental in understanding that vehicle handling is not a static property but a dynamic state influenced by the driver's use of the pedals. The Steering Pad test confirmed the theoretical drop of the rear axle characteristic curve under traction, while the wet handling session provided a tangible sense of the importance of tire engineering in ensuring vehicle stability at the limit of adhesion.

List of Figures

1.1	Model of vehicle considered as point	1
1.2	Lamborghini Miura P400S	8
1.3	Engine torque and power map of the Lamborghini Miura P400S	11
1.4	Tractive and resistive power for the fifth gear of the Lamborghini Miura P400S	13
1.5	Tractive and resistive power at the first gear for maximum incline conditions	14
1.6	Tractive and resistive power at the first gear for maximum incline conditions	14
1.7	Tractive power on speed and Speed on RPMs for the all the gears of the Lamborghini Miura P400S	15
1.8	Tractive and resistive power for the fifth gear of the Lamborghini Miura P400S	19
1.9	Tractive and resistive power at the first gear for maximum incline conditions	19
1.10	Tractive and resistive power at the first gear for maximum incline conditions	20
1.11	Tractive power on speed and Speed on RPMs for the all the gears of the Lamborghini Miura P400S	21
2.1	Quarter car model scheme.	23
2.2	Sensitivity for Alfa Romeo Alfetta.	25
3.1	Vehicle architecture.	27
3.2	Vehicle architecture.	29
4.1	Motorcycle scheme.	31
4.2	$\mu_r - \mu_f$ diagram.	34
4.3	$F_{x,f} - F_{x,r}$ diagram.	34
4.4	Deceleration limits.	35
5.1	Example of MacPherson strut	40
5.2	Main kinematic points of a MacPherson suspension	41
5.3	Matlab SimMechanics model of the suspension	43

5.4	Camber (left) and steering (right) variation as function of the stroke after kinematic optimization. The blue line is the output computed by the SimMechanics model, the red circles are the target values.	44
5.5	Steering angles of the wheel	47
5.6	Stress Distribution of the Lower Arm	49
5.7	Displacements of the Lower Arm	49
5.8	Suspension Assembly	50
5.9	Suspension Assembly - Front View	51
6.1	Sines stresses distribution on the rim.	55
6.2	Number of admissible cycles per point.	55
7.1	Vertical accelerations experienced during test.. . . .	58
7.2	Truncated signal, with tests at different excitation frequencies.	59
7.3	Spectrum of the signal for each signal section.	59
7.4	Power on manouver	67
7.5	Pirelli test track map.	68

List of Tables

1.1	Powertrain Parameters	10
1.2	Vehicle Parameters	10
1.3	Calculated and actual gear ratios of the Lamborghini Miura P400S	15
1.4	Gear Ratios in Full-Load and Running Order Condition compared to real ones	20
2.1	Cars data for quarter vehicle model	23
2.2	σ_{x2} for normal and rough roads	24
2.3	Parameters' effect on σ_{x2}	25
3.1	Vehicle data	28
3.2	Overall consumption - Case 1	29
3.3	Overall consumption - Case 2	30
4.1	Vehicle and Rider Specifications	31
5.1	Main kinematic points	42
5.2	Target values of camber and steer angle	42
5.3	Optimal value of the kinematic points by curve fitting	43
5.4	Mechanical properties of 6061-T6 aluminium alloy	46
5.5	Summary of FEA results	48

List of Symbols

Variable	Description	SI unit
$\boldsymbol{u}$	solid displacement	m
$\boldsymbol{u}_f$	fluid displacement	m